

Lab 1: Network & Service Reconnaissance with Metasploitable 2

Student Name: Maryjudith Chidinma Ogunaka

Registration Number: C11/26/FCDF/17151

Programme: ICDFA Trainee | Cohort 11

Lab Title: Lab 1: Network & Service Reconnaissance with Metasploitable 2

Date: 05 September 2026

Authorised Target: Metasploitable 2 Virtual Machine

Initial Target IP Address (Steps 1-9): 10.15.203.233

Updated Target IP Address (Steps 10-30): 10.15.203.91

Executive Summary

This lab focused on network reconnaissance and service enumeration using Nmap and WhatWeb on the Metasploitable 2 virtual machine within an authorized lab environment. The objective was to identify active hosts, discover open ports, determine running services, detect operating system information, and fingerprint web technologies.

Several scanning techniques were performed, including host discovery, TCP and UDP port scanning, service version detection, operating system detection, NSE script enumeration, and web fingerprinting. The assessment identified multiple services running on the target system, including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, MySQL, NFS, VNC, IRC, and Apache Tomcat.

WhatWeb fingerprinting revealed that the web application was running on Apache 2.2.8, Ubuntu Linux, and PHP 5.2.4-2ubuntu5.10 with WebDAV enabled. The information gathered provided a comprehensive inventory of the target's exposed services and technologies.

The lab successfully demonstrated how reconnaissance tools can be used to gather information about a target environment and highlighted the importance of information gathering in cybersecurity assessments.

Objective

The objectives of this laboratory exercise were to:

- Identify the IP addresses of the Kali Linux scanner and the Metasploitable 2 target system and verify network connectivity between both hosts.

- Perform host discovery to confirm that the target system was active and reachable on the network.
- Conduct default, service version detection, operating system detection, aggressive, full TCP, and UDP scans using Nmap.
- Utilize Nmap Scripting Engine (NSE) scripts to enumerate HTTP, SMB, SSH, and FTP services and gather additional service information.
- Fingerprint the target web application using WhatWeb at different aggression levels to identify web technologies and server components.
- Create a comprehensive service inventory based on the information gathered during reconnaissance and enumeration activities.
- Analyze and document all findings and provide answers to the conceptual questions associated with the laboratory exercise.

Lab Environment and Connectivity

The laboratory exercise was conducted in an isolated and authorized virtual environment consisting of **Kali Linux** as the scanning and assessment machine and **Metasploitable 2** as the target system. Both virtual machines were connected to the same network segment to allow communication and reconnaissance activities.

The Metasploitable 2 target system was initially identified using the `ifconfig` command and assigned the IP address **10.15.203.233**. The Kali Linux workstation was identified using the `ip addr` command and confirmed to be operating on the same subnet. A four-packet ICMP ping test was successfully performed to verify connectivity between both systems before scanning activities commenced.

All activities from **Step 1 to Step 9** were conducted using the target IP address **10.15.203.233**. During the exercise, the Metasploitable 2 virtual machine was restarted and subsequently received a new DHCP-assigned IP address, **10.15.203.91**. The address change was verified through a second `ifconfig` check on the target system.

As a result, all activities from **Step 10 to Step 30** were performed using the updated IP address **10.15.203.91**. Although two different IP addresses were used during the assessment, both addresses belonged to the same Metasploitable 2 virtual machine. Therefore, all findings remained valid and were consolidated into a single report. Connectivity between Kali Linux and Metasploitable 2 was maintained throughout the exercise, enabling the successful completion of all reconnaissance, enumeration, and web fingerprinting tasks.

Methodology

The assessment was conducted using Kali Linux against the authorized Metasploitable 2 target machine in an isolated laboratory environment. Reconnaissance activities were performed sequentially, beginning with host discovery and connectivity verification. Nmap was then used to conduct port scanning, service version detection, operating system fingerprinting, aggressive scanning, full TCP and UDP scans, and NSE-based service enumeration. Additional reconnaissance was performed using HTTP enumeration techniques and WhatWeb fingerprinting to identify web technologies and server components. All activities were conducted solely against the authorized lab target.

Step 1: Identify Metasploitable 2 IP Address

[illegible]

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:ae:9a:6c
          inet addr:10.15.203.233  Bcast:10.15.203.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:feae:9a6c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:37 errors:0 dropped:0 overruns:0 frame:0
          TX packets:66 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4530 (4.4 KB)  TX bytes:6968 (6.8 KB)
          Base address:0x2000 Memory:fd5c0000-fd5e0000

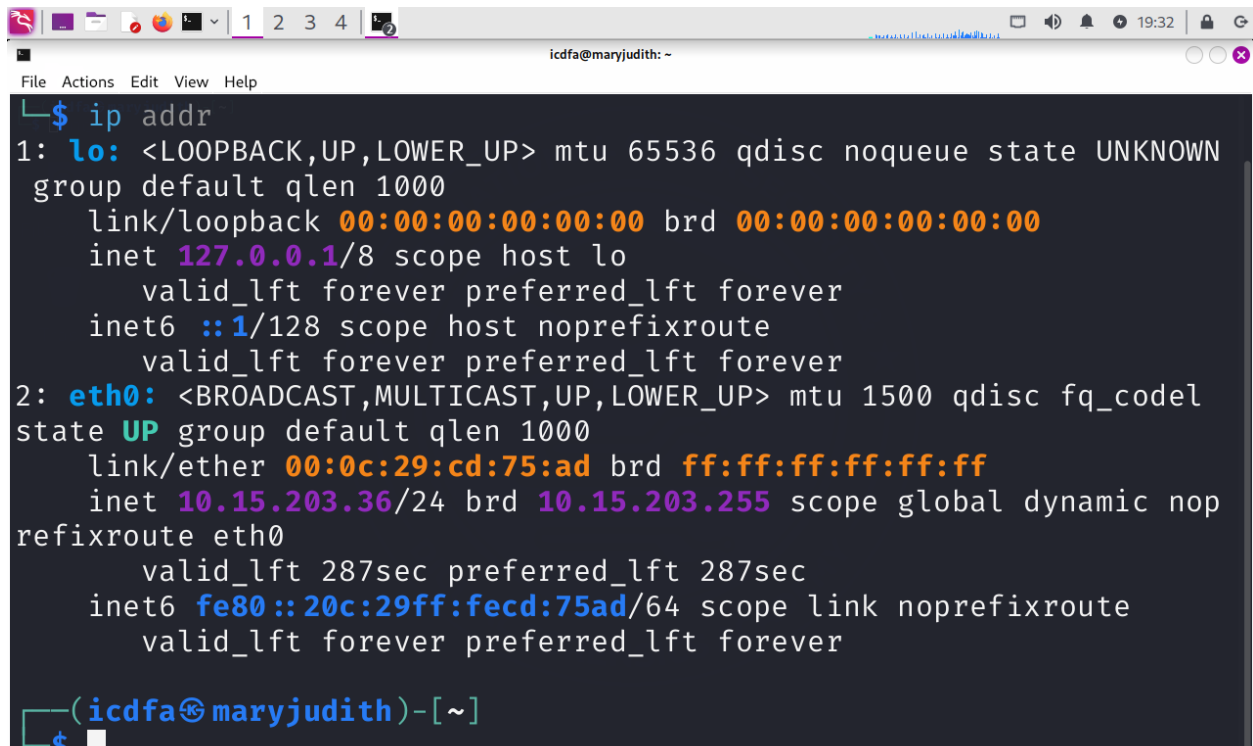
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$ m
```

Report :

The ifconfig command was executed on the Metasploitable 2 virtual machine to identify its network configuration and assigned IP address. The output displayed the active network interface (eth0) and its corresponding IP address. The target system was assigned the IP address **10.15.203.233**, which was recorded and used for subsequent reconnaissance and scanning activities. Identifying the target IP address was a necessary first step because all communication and security assessment activities required a valid destination address. This confirmed that the Metasploitable 2 machine was connected to the network and ready for testing.

Step 2: Identify Kali Linux IP Address



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN  
   group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel  
   state UP group default qlen 1000  
    link/ether 00:0c:29:cd:75:ad brd ff:ff:ff:ff:ff:ff  
    inet 10.15.203.36/24 brd 10.15.203.255 scope global dynamic nop  
refixroute eth0  
        valid_lft 287sec preferred_lft 287sec  
    inet6 fe80::20c:29ff:fe75:ad/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(icdfa@maryjudith)-[~]  
$
```

Report

The ip addr command was executed on the Kali Linux virtual machine to determine its network configuration and assigned IP address. The output displayed the active network interface and its associated IP address. This information was used to confirm that the Kali Linux system was connected to the same network as the Metasploitable 2 target machine. Verifying the scanner's IP address ensured that communication between the two virtual machines could be established successfully. The identified IP address was subsequently used as the source system for reconnaissance and scanning activities.

Step 3: Connectivity Verification

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
refixroute eth0  
    valid_lft 287sec preferred_lft 287sec  
    inet6 fe80::20c:29ff:fe8d:75ad/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
  
(icdfa@maryjudith)-[~]  
$ ping -c 4 10.15.203.233  
PING 10.15.203.233 (10.15.203.233) 56(84) bytes of data.  
64 bytes from 10.15.203.233: icmp_seq=1 ttl=64 time=1.56 ms  
64 bytes from 10.15.203.233: icmp_seq=2 ttl=64 time=1.87 ms  
64 bytes from 10.15.203.233: icmp_seq=3 ttl=64 time=0.763 ms  
64 bytes from 10.15.203.233: icmp_seq=4 ttl=64 time=1.56 ms  
  
— 10.15.203.233 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3006ms  
rtt min/avg/max/mdev = 0.763/1.439/1.873/0.410 ms  
  
(icdfa@maryjudith)-[~]  
$
```

Report

A connectivity test was performed using the ping command to verify communication between the Kali Linux virtual machine and the Metasploitable 2 target system. Four ICMP Echo Request packets were sent to the target host, and successful responses were received. The results confirmed that the target system was online, reachable, and connected to the same network as the Kali Linux machine. This step was important because it verified network connectivity before proceeding with reconnaissance and scanning activities.

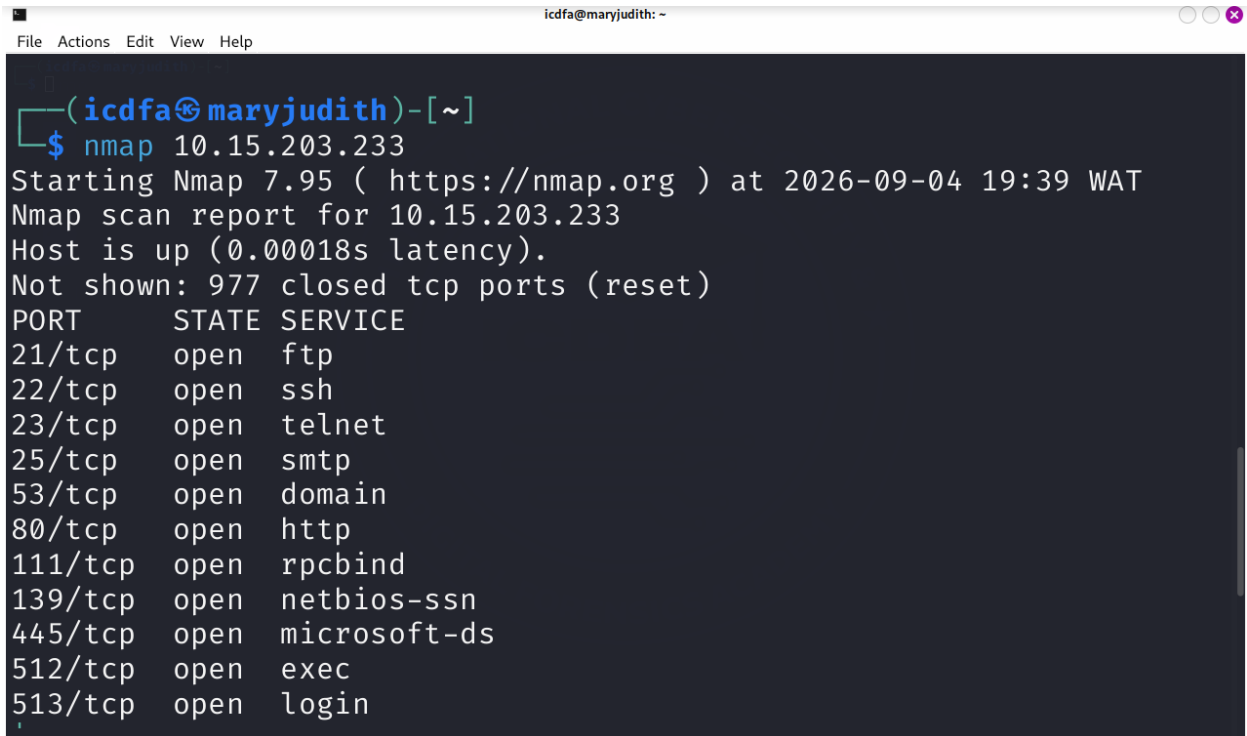
Step 4: Host Discovery Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ nmap -sn 10.15.203.0/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:36 WAT  
Nmap scan report for 10.15.203.122  
Host is up (0.024s latency).  
MAC Address: B6:56:F3:DF:51:A5 (Unknown)  
Nmap scan report for 10.15.203.193  
Host is up (0.00036s latency).  
MAC Address: 90:CC:DF:A7:CD:12 (Intel Corporate)  
Nmap scan report for 10.15.203.217  
Host is up (0.058s latency).  
MAC Address: AA:30:8E:F8:A3:D6 (Unknown)  
Nmap scan report for 10.15.203.233  
Host is up (0.00078s latency).  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Nmap scan report for 10.15.203.36  
Host is up.  
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.94 seconds  
  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:36 WAT  
Nmap scan report for 10.15.203.122  
Host is up (0.024s latency).  
MAC Address: B6:56:F3:DF:51:A5 (Unknown)  
Nmap scan report for 10.15.203.193  
Host is up (0.00036s latency).  
MAC Address: 90:CC:DF:A7:CD:12 (Intel Corporate)  
Nmap scan report for 10.15.203.217  
Host is up (0.058s latency).  
MAC Address: AA:30:8E:F8:A3:D6 (Unknown)  
Nmap scan report for 10.15.203.233  
Host is up (0.00078s latency).  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Nmap scan report for 10.15.203.36  
Host is up.  
Nmap done: 256 IP addresses (5 hosts up) scanned in 7.94 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

A host discovery scan was conducted using the `nmap -sn` command to identify active devices on the network. The scan examined hosts within the specified network range and successfully detected multiple live systems. The Metasploitable 2 target machine was identified among the active hosts, confirming that it was online and reachable. This step helped verify the presence of the target before proceeding with more detailed reconnaissance and service enumeration activities.

Step 5: Default TCP Scan



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ nmap 10.15.203.233  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:39 WAT  
Nmap scan report for 10.15.203.233  
Host is up (0.00018s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
513/tcp open login  
514/tcp open shell  
1099/tcp open rmiregistry  
1524/tcp open ingreslock  
2049/tcp open nfs  
2121/tcp open ccproxy-ftp  
3306/tcp open mysql  
5432/tcp open postgresql  
5900/tcp open vnc  
6000/tcp open X11  
6667/tcp open irc  
8009/tcp open ajp13  
8180/tcp open unknown  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

A default Nmap scan was performed against the Metasploitable 2 target system to identify open TCP ports and running services. The scan examined the most commonly used ports and revealed multiple open services available on the target machine. The results provided an initial overview of the target's network exposure and identified potential points of further investigation. Information gathered during this stage was used as a foundation for subsequent service enumeration and version detection activities. The scan confirmed that the target was hosting several network services and was actively responding to connection requests.

Step 6: Service Version Detection

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ nmap -sV 10.15.203.233  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:41 WAT  
Nmap scan report for 10.15.203.233  
Host is up (0.00019s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol  
2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGRO
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
UP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
512/tcp   open  exec         netkit-rsh rexecd  
513/tcp   open  login  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitabl
```

```
File Actions Edit View Help
2049/tcp open  nfs      2-4 (RPC #100003)
2121/tcp open  ftp      ProFTPD 1.3.1
3306/tcp open  mysql    MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc      VNC (protocol 3.3)
6000/tcp open  X11      (access denied)
6667/tcp open  irc      UnrealIRCd
8009/tcp open  ajp13    Apache Jserv (Protocol v1.3)
8180/tcp open  http     Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:AE:9A:6C (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.67 seconds

(icdfa@maryjudith)-[~]
$
```

Report

A service version detection scan was performed using the `-sV` option to identify the services and software versions running on the target system. The scan successfully detected multiple services and provided detailed information about the applications operating on the open ports. This information improved visibility into the target environment and helped identify the technologies deployed on the system. Service version detection is an important reconnaissance technique because it allows analysts to determine the exact software running on network services. The results obtained from this scan contributed to the creation of a detailed service inventory for the target machine.

Step 7: Maximum Version Intensity Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
icdfa@maryjudith)-[~]  
$ nmap -sV --version-intensity 9 10.15.203.233  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:46 WAT  
Nmap scan report for 10.15.203.233  
Host is up (0.00017s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp           vsftpd 2.3.4  
22/tcp    open  ssh           OpenSSH 4.7p1 Debian 8ubuntu1 (protocol  
2.0)  
23/tcp    open  telnet        Linux telnetd  
25/tcp    open  smtp          Postfix smtpd  
53/tcp    open  domain        ISC BIND 9.4.2  
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind       2 (RPC #100000)  
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
512/tcp   open  exec          netkit-rsh rexecd  
513/tcp   open  login         OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi      GNU Classpath grmiregistry  
1524/tcp  open  bindshell     Metasploitable root shell  
2049/tcp  open  nfs           2-4 (RPC #100003)  
2121/tcp  open  ftp           ProFTPD 1.3.1  
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc           VNC (protocol 3.3)  
6000/tcp  open  X11           (access denied)  
6667/tcp  open  irc           UnrealIRCd  
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)  
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitabl  
e.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
2049/tcp open  nfs      2-4 (RPC #100003)  
2121/tcp open  ftp      ProFTPD 1.3.1  
3306/tcp open  mysql    MySQL 5.0.51a-3ubuntu5  
5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp open  vnc      VNC (protocol 3.3)  
6000/tcp open  X11      (access denied)  
6667/tcp open  irc      UnrealIRCd  
8009/tcp open  ajp13    Apache Jserv (Protocol v1.3)  
8180/tcp open  http     Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 11.55 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

A detailed service version detection scan was performed using the `--version-intensity 9` option to increase the accuracy of service identification on the target system. This scan used additional probes to gather more information about the services running on open ports. The results provided more detailed service information and improved confidence in the identified software versions. This approach allowed a more thorough assessment of the target environment and supported the validation of previous service detection results. The information obtained contributed to a more complete understanding of the services available on the Metasploitable 2 system.

Step 8: Operating System Detection

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ sudo nmap -O 10.15.203.233  
sudo: unable to resolve host maryjudith: Name or service not known  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:51 WAT  
Nmap scan report for 10.15.203.233  
Host is up (0.0016s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Device type: general purpose
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
5432/tcp open  postgresql  
5900/tcp open  vnc  
6000/tcp open  X11  
6667/tcp open  irc  
8009/tcp open  ajp13  
8180/tcp open  unknown  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at http://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.58 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

An operating system detection scan was conducted using the -O option to identify the operating system running on the target machine. The scan successfully confirmed that the target host was active and revealed multiple open TCP ports, including FTP, SSH, Telnet, SMTP, DNS, HTTP, RPC, NetBIOS, and SMB services. Nmap analyzed responses from the target system to determine characteristics of the underlying operating system. The results provided valuable information about the target environment and supported further reconnaissance activities. Identifying the operating system helps security analysts understand the platform being assessed and improves subsequent enumeration efforts.

Step 9: Aggressive Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ sudo nmap -A 10.15.203.233  
sudo: unable to resolve host maryjudith: Name or service not known  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 19:55 WAT  
Nmap scan report for 10.15.203.233  
Host is up (0.00085s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
| ftp-syst:  
|   STAT:  
| FTP server status:  
|   Connected to 10.15.203.36  
|   Logged in as ftp  
|   TYPE: ASCII  
|   No session bandwidth limit  
|   Session timeout in seconds is 300  
|   Control connection is plain text
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|   Control connection is plain text  
|   Data connections will be plain text  
|   vsFTPD 2.3.4 - secure, fast, stable  
|_End of status  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol  
2.0)  
| ssh-hostkey:  
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
|_ssl-date: 2026-09-05T02:08:44+00:00; +7h13m14s from scanner time.  
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organiza  
tionName=OCOSA/stateOrProvinceName=There is no such thing outside U  
S/countryName=XX  
| Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_Not valid after: 2010-04-16T14:07:45  
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240  
000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN  
| sslv2:  
|   SSLv2 supported  
|   ciphers:  
|     SSL2_RC4_128_EXPORT40_WITH_MD5  
|     SSL2_DES_64_CBC_WITH_MD5  
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5  
|     SSL2_RC2_128_CBC_WITH_MD5  
|     SSL2_DES_192_EDE3_CBC_WITH_MD5  
|     SSL2_RC4_128_WITH_MD5  
|_ 53/tcp open domain ISC BIND 9.4.2  
| dns-nsid:  
|_ bind.version: 9.4.2  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_http-title: Metasploitable2 - Linux  
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
111/tcp open rpcbind 2 (RPC #100000)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_http-title: Metasploitable2 - Linux  
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
111/tcp open rpcbind 2 (RPC #100000)  
| rpcinfo:  
|   program version port/proto service  
|   100000 2 111/tcp rpcbind  
|   100000 2 111/udp rpcbind  
|   100003 2,3,4 2049/tcp nfs  
|   100003 2,3,4 2049/udp nfs  
|   100005 1,2,3 44260/udp mountd  
|   100005 1,2,3 60658/tcp mountd  
|   100021 1,3,4 34641/tcp nlockmgr  
|   100021 1,3,4 38535/udp nlockmgr  
|   100024 1 35839/tcp status  
|   100024 1 39614/udp status  
|_ 139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WOR  
KGROUP)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_ 100024 1 39614/udp status  
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WOR  
KGROUP)  
512/tcp open exec netkit-rsh rexecd  
513/tcp open login OpenBSD or Solaris rlogind  
514/tcp open tcpwrapped  
1099/tcp open java-rmi GNU Classpath grmiregistry  
1524/tcp open bindshell Metasploitable root shell  
2049/tcp open nfs 2-4 (RPC #100003)  
2121/tcp open ftp ProFTPD 1.3.1  
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5  
| mysql-info:  
| Protocol: 10  
| Version: 5.0.51a-3ubuntu5  
| Thread ID: 10  
| Capabilities flags: 43564  
| Some Capabilities: SwitchToSSLAfterHandshake, Support41Auth, Su
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
| Some Capabilities: SwitchToSSLAfterHandshake, Support41Auth, Su  
portsCompression, SupportsTransactions, Speaks41ProtocolNew, LongC  
olumnFlag, ConnectWithDatabase  
| Status: Autocommit  
|_ Salt: \N2Mm`vQri8z~*UMc,L  
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7  
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organiza  
tionName=OCOSA/stateOrProvinceName=There is no such thing outside U  
S/countryName=XX  
| Not valid before: 2010-03-17T14:07:45  
|_ Not valid after: 2010-04-16T14:07:45  
|_ ssl-date: 2026-09-05T02:08:45+00:00; +7h13m14s from scanner time.  
5900/tcp open vnc VNC (protocol 3.3)  
| vnc-info:  
| Protocol version: 3.3  
| Security types:  
|_ VNC Authentication (2)  
6000/tcp open X11 (access denied)  
6667/tcp open irc UnrealIRCd
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_ VNC Authentication (2)  
6000/tcp open  X11          (access denied)  
6667/tcp open  irc            UnrealIRCd  
| irc-info:  
|   users: 1  
|   servers: 1  
|   lusers: 1  
|   lservers: 0  
|   server: irc.Metasploitable.LAN  
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN  
|   uptime: 0 days, 0:26:27  
|   source ident: nmap  
|   source host: D7F09409.9BC63D70.BC17435D.IP  
|_ error: Closing Link: izkfwlalh[10.15.203.36] (Quit: izkfwlalh)  
8009/tcp open  ajp13         Apache Jserv (Protocol v1.3)  
|_ajp-methods: Failed to get a valid response for the OPTION request  
t  
8180/tcp open  http          Apache Tomcat/Coyote JSP engine 1.1  
|_http-title: Apache Tomcat/5.5
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
t  
8180/tcp open  http          Apache Tomcat/Coyote JSP engine 1.1  
|_http-title: Apache Tomcat/5.5  
|_http-favicon: Apache Tomcat  
|_http-server-header: Apache-Coyote/1.1  
MAC Address: 00:0C:29:AE:9A:6C (VMware)  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Host script results:  
|_smb2-time: Protocol negotiation failed (SMB2)  
|_clock-skew: mean: 8h13m14s, deviation: 2h00m00s, median: 7h13m13s  
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
```

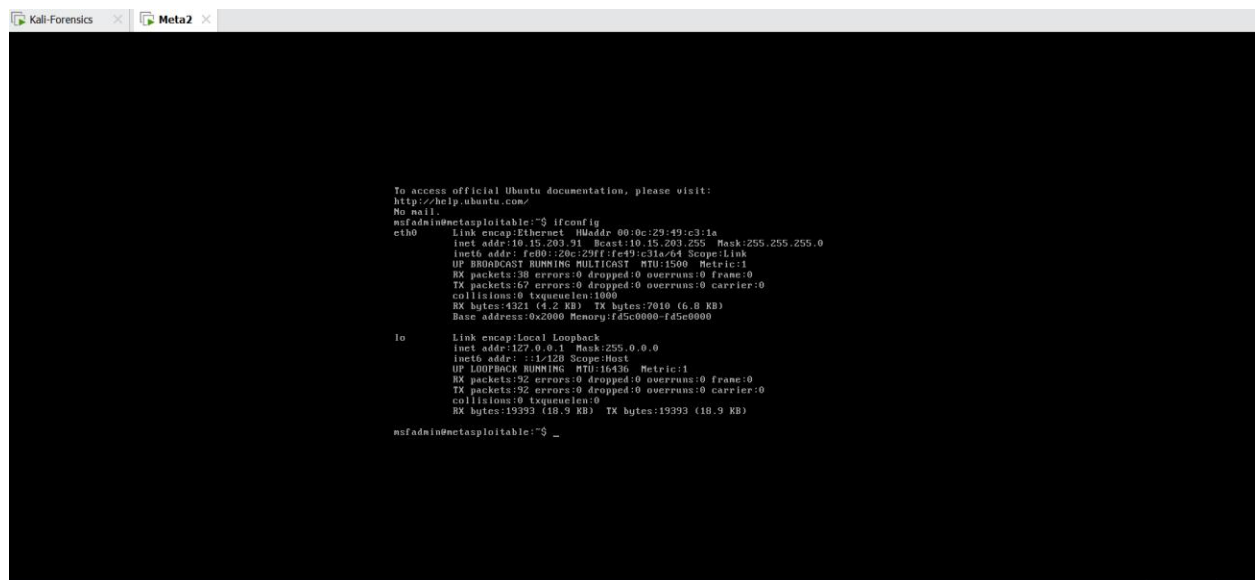
```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_smb2-time: Protocol negotiation failed (SMB2)  
|_clock-skew: mean: 8h13m14s, deviation: 2h00m00s, median: 7h13m13s  
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)  
| smb-security-mode:  
|   account_used: <blank>  
|   authentication_level: user  
|   challenge_response: supported  
|_ message_signing: disabled (dangerous, but default)  
| smb-os-discovery:  
|   OS: Unix (Samba 3.0.20-Debian)  
|   Computer name: metasploitable  
|   NetBIOS computer name:  
|   Domain name: localdomain  
|   FQDN: metasploitable.localdomain  
|_ System time: 2026-09-04T22:08:35-04:00  
  
TRACEROUTE  
HOP RTT ADDRESS
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_ message_signing: disabled (dangerous, but default)  
| smb-os-discovery:  
|   OS: Unix (Samba 3.0.20-Debian)  
|   Computer name: metasploitable  
|   NetBIOS computer name:  
|   Domain name: localdomain  
|   FQDN: metasploitable.localdomain  
|_ System time: 2026-09-04T22:08:35-04:00  
  
TRACEROUTE  
HOP RTT ADDRESS  
1 0.85 ms 10.15.203.233  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 23.46 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

An aggressive scan was performed using the -A option to gather detailed information about the target system. This scan combined operating system detection, service version detection, default NSE scripts, and traceroute functionality into a single operation. The results provided detailed information about open ports, running services, service versions, and host configuration details. Additional information obtained from NSE scripts helped improve visibility into the target environment. The aggressive scan produced a comprehensive profile of the target system and assisted in developing a detailed service inventory.

IP Address Change Observation



```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:49:c3:1a
          inet addr:10.15.203.91  Bcast:10.15.203.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe49:c31a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:38  errors:0  dropped:0  overruns:0  frame:0
          TX packets:67  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:1000
          RX bytes:4321 (4.2 KB)  TX bytes:7010 (6.8 KB)
          Base address:0x2000  Memory:f45c0000-f45c0000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1:1:1 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:52  errors:0  dropped:0  overruns:0  frame:0
          TX packets:52  errors:0  dropped:0  overruns:0  carrier:0
          collisions:0  txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ _
```

During the assessment, the Metasploitable 2 target system received a new IP address due to a virtual machine network change. The target IP address changed from 10.15.203.233 to 10.15.203.91, which was confirmed using the ifconfig command. After validating the new address, all subsequent reconnaissance and scanning activities were directed to 10.15.203.91. The results collected before the IP change remained valid because both IP addresses belonged to the same target system.

Note: Verification of the Metasploitable 2 IP address change, with 10.15.203.91 used as the target address from Step 10 through Step 30.

Step 10: Full TCP Port Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
(icdfa@maryjudith)-[~]  
$ sudo nmap -p- 10.15.203.91  
sudo: unable to resolve host maryjudith: Name or service not known  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 18:02 WAT  
Stats: 0:03:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 68.61% done; ETC: 18:07 (0:01:29 rem  
aining)  
Stats: 0:06:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 70.00% done; ETC: 18:12 (0:02:49 rem  
aining)  
Stats: 0:12:05 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 72.33% done; ETC: 18:19 (0:04:37 rem  
aining)  
Stats: 0:18:35 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 75.06% done; ETC: 18:27 (0:06:11 rem
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
SYN Stealth Scan Timing: About 82.00% done; ETC: 18:45 (0:07:42 rem  
aining)  
Stats: 0:35:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 82.14% done; ETC: 18:46 (0:07:42 rem  
aining)  
Stats: 0:37:35 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 83.06% done; ETC: 18:48 (0:07:40 rem  
aining)  
Stats: 0:37:37 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 83.07% done; ETC: 18:48 (0:07:40 rem  
aining)  
Stats: 0:37:38 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 83.08% done; ETC: 18:48 (0:07:40 rem  
aining)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
SYN Stealth Scan Timing: About 87.02% done; ETC: 18:57 (0:07:04 remaining)  
Stats: 0:47:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 87.03% done; ETC: 18:57 (0:07:04 remaining)  
Stats: 0:50:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 88.16% done; ETC: 18:59 (0:06:44 remaining)  
Stats: 0:50:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 88.16% done; ETC: 18:59 (0:06:44 remaining)  
Stats: 0:58:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 91.90% done; ETC: 19:07 (0:05:12 remaining)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
Stealth Scan  
SYN Stealth Scan Timing: About 98.91% done; ETC: 19:19 (0:00:50 remaining)  
Stats: 1:17:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 99.74% done; ETC: 19:20 (0:00:12 remaining)  
Nmap scan report for 10.15.203.91  
Host is up (0.00019s latency).  
Not shown: 43502 closed tcp ports (reset), 22013 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
1099/tcp  open  rmiregistry  
2049/tcp  open  nfs  
3306/tcp  open  mysql  
3632/tcp  open  distccd  
5900/tcp  open  vnc  
6667/tcp  open  irc  
8787/tcp  open  msgsrvr
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
139/tcp    open  netbios-ssn  
445/tcp    open  microsoft-ds  
512/tcp    open  exec  
513/tcp    open  login  
1099/tcp   open  rmiregistry  
2049/tcp   open  nfs  
3306/tcp   open  mysql  
3632/tcp   open  distccd  
5900/tcp   open  vnc  
6667/tcp   open  irc  
8787/tcp   open  msgsrvr  
48708/tcp  open  unknown  
57785/tcp  open  unknown  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 4693.42 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

During the practical exercise, the target IP address changed due to the virtual machine network configuration. Initially, the Metasploitable 2 target was assigned the IP address 10.15.203.233. However, after restarting or reconnecting the virtual machine, the target received a new IP address of 10.15.203.91. The change was verified using the `ifconfig` command on the Metasploitable 2 system. Since both IP addresses belonged to the same target machine, the results obtained from previous scans remained valid and were retained for analysis. All subsequent scanning and reconnaissance activities were performed using the updated IP address 10.15.203.91.

Step 11: Full TCP Scan with Version Detection

```
icdta@maryjudith: ~  
File Actions Edit View Help  
(icdfa@maryjudith)-[~]  
$ sudo nmap -p- -sV 10.15.203.91  
sudo: unable to resolve host maryjudith: Name or service not known  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 20:26 WAT  
Stats: 0:01:44 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan  
Service scan Timing: About 96.67% done; ETC: 20:28 (0:00:03 remaining)  
Nmap scan report for 10.15.203.91  
Host is up (0.00057s latency).  
Not shown: 65505 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
```

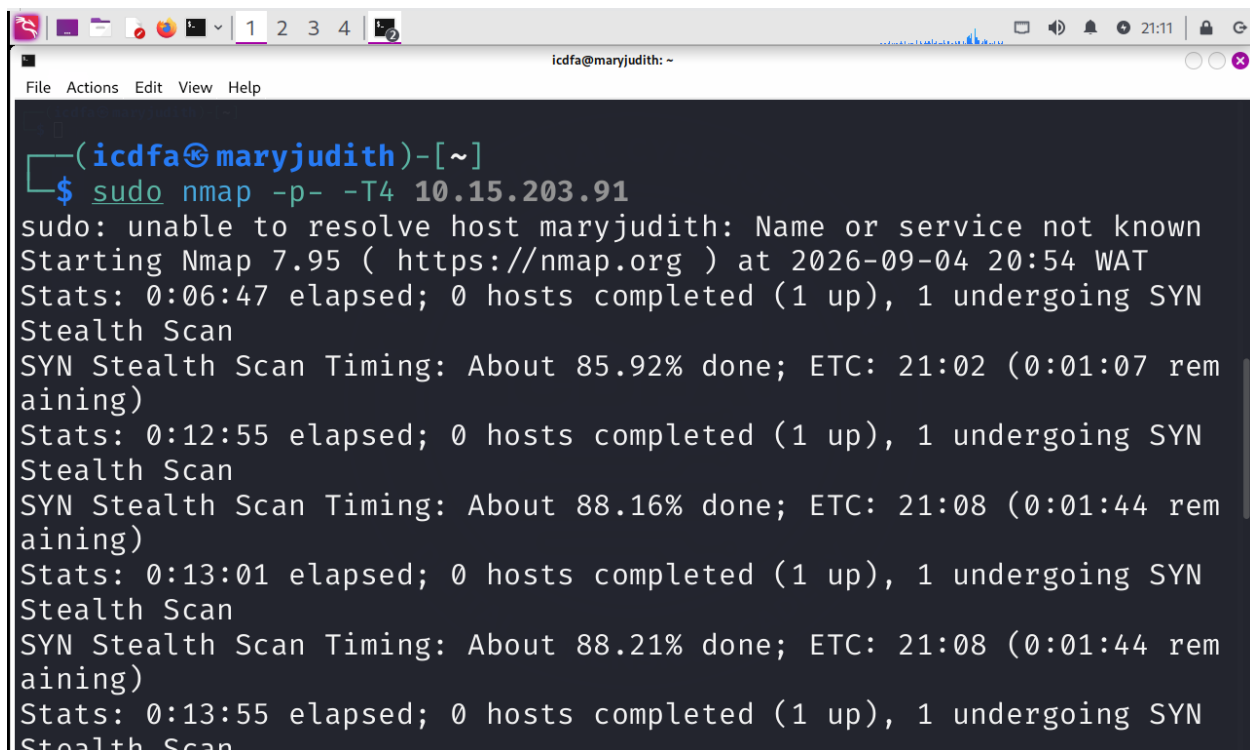
```
icdfa@maryjudith: ~  
File Actions Edit View Help  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp open rpcbind 2 (RPC #100000)  
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGR  
OUP)  
445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGR  
OUP)  
512/tcp open exec netkit-rsh rexecd  
513/tcp open login  
514/tcp open tcpwrapped  
1099/tcp open java-rmi GNU Classpath grmiregistry  
1524/tcp open bindshell Metasploitable root shell  
2049/tcp open nfs 2-4 (RPC #100003)  
2121/tcp open ftp ProFTPD 1.3.1  
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5  
3632/tcp open distccd distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1  
ubuntu4))  
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp open vnc VNC (protocol 3.3)  
6000/tcp open X11 (access denied)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
6000/tcp open X11 (access denied)  
6667/tcp open irc UnrealIRCd  
6697/tcp open irc UnrealIRCd  
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)  
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1  
8787/tcp open drb Ruby DRb RMI (Ruby 1.8; path /usr/lib/r  
uby/1.8/drb)  
37084/tcp open java-rmi GNU Classpath grmiregistry  
41897/tcp open mountd 1-3 (RPC #100005)  
50746/tcp open status 1 (RPC #100024)  
54630/tcp open nlockmgr 1-4 (RPC #100021)  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitabl  
e.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 131.82 seconds
```

Report

A full TCP scan with version detection was performed using `sudo nmap -p- -sV 10.15.203.233` to identify all open TCP ports and determine the services and software versions running on the target system. This scan combined comprehensive port enumeration with service version identification to provide a detailed inventory of services available on the Metasploitable 2 machine. By examining all 65,535 TCP ports, the scan was able to detect both common and less frequently used services. The version detection feature revealed information about the applications running on open ports and the software versions associated with those services. This information is valuable because it provides greater visibility into the target environment and assists in identifying technologies deployed on the system. The results contributed significantly to the creation of a comprehensive service inventory and supported further analysis of the target machine.

Step 12: Timing Template Scan

A screenshot of a terminal window titled 'icdfa@maryjudith: ~'. The terminal shows a command prompt where the user has entered 'sudo nmap -p- -T4 10.15.203.91'. The output of the command is displayed in a dark-themed terminal. It shows the start of an Nmap SYN Stealth Scan on host 10.15.203.91. The output includes the Nmap version (7.95), the start time (2026-09-04 20:54 WAT), and progress updates. The scan is currently at 88.21% completion, with 0 hosts completed and 1 host undergoing a SYN Stealth Scan. The estimated time remaining is 0:01:44.

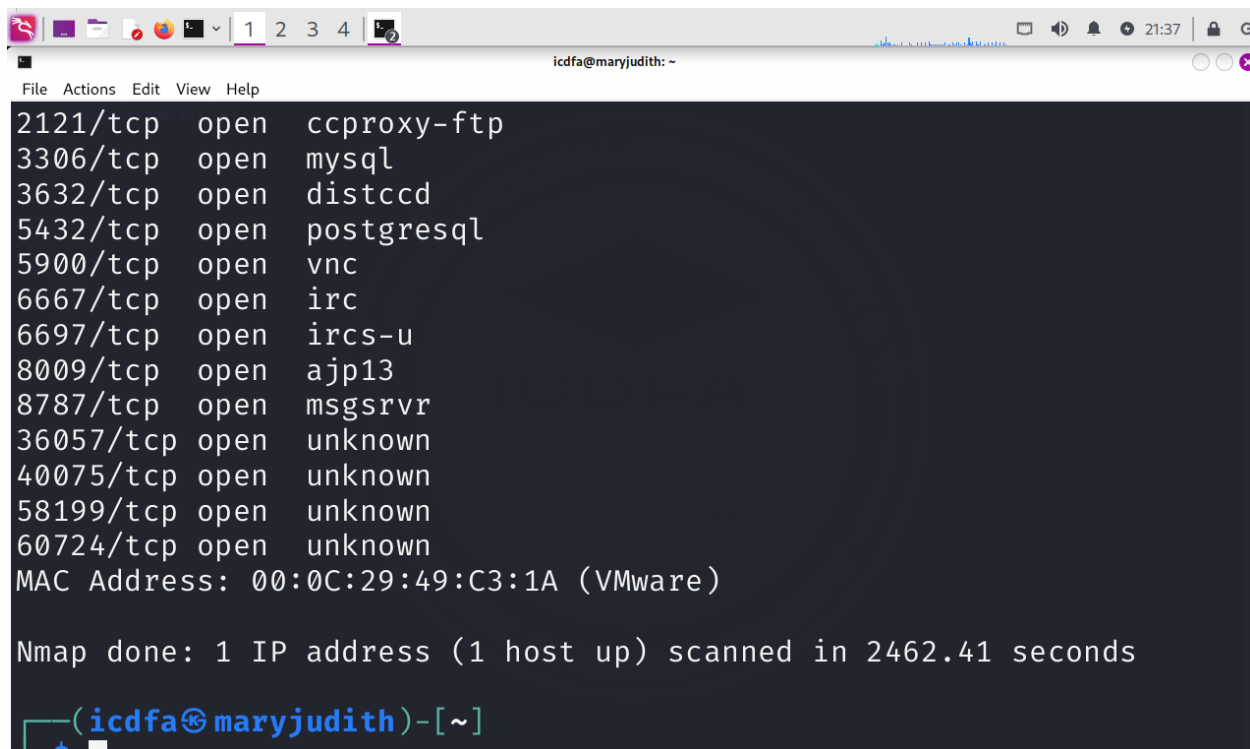
```
(icdfa@maryjudith)-[~]
$ sudo nmap -p- -T4 10.15.203.91
sudo: unable to resolve host maryjudith: Name or service not known
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 20:54 WAT
Stats: 0:06:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN
Stealth Scan
SYN Stealth Scan Timing: About 85.92% done; ETC: 21:02 (0:01:07 rem
aining)
Stats: 0:12:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN
Stealth Scan
SYN Stealth Scan Timing: About 88.16% done; ETC: 21:08 (0:01:44 rem
aining)
Stats: 0:13:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN
Stealth Scan
SYN Stealth Scan Timing: About 88.21% done; ETC: 21:08 (0:01:44 rem
aining)
Stats: 0:13:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN
Stealth Scan
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
SYN Stealth Scan Timing: About 88.21% done; ETC: 21:08 (0:01:44 remaining)  
Stats: 0:13:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 88.59% done; ETC: 21:09 (0:01:48 remaining)  
Stats: 0:13:56 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 88.60% done; ETC: 21:09 (0:01:48 remaining)  
Stats: 0:16:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 89.54% done; ETC: 21:12 (0:01:53 remaining)  
Stats: 0:17:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 89.91% done; ETC: 21:13 (0:01:55 remaining)  
Stats: 0:17:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
SYN Stealth Scan Timing: About 89.93% done; ETC: 21:13 (0:01:55 remaining)  
Stats: 0:17:08 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 89.94% done; ETC: 21:13 (0:01:55 remaining)  
Stats: 0:17:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 89.95% done; ETC: 21:13 (0:01:55 remaining)  
Stats: 0:17:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 89.96% done; ETC: 21:13 (0:01:55 remaining)  
Stats: 0:23:50 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 92.76% done; ETC: 21:19 (0:01:52 remaining)
```

```
icdfa@maryjudith: ~  
Stats: 0:38:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 98.74% done; ETC: 21:32 (0:00:29 rem  
aining)  
Stats: 0:39:57 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 99.55% done; ETC: 21:34 (0:00:11 rem  
aining)  
Stats: 0:40:33 elapsed; 0 hosts completed (1 up), 1 undergoing SYN  
Stealth Scan  
SYN Stealth Scan Timing: About 99.80% done; ETC: 21:34 (0:00:05 rem  
aining)  
Nmap scan report for 10.15.203.91  
Host is up (0.00019s latency).  
Not shown: 54020 closed tcp ports (reset), 11487 filtered tcp ports  
(no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
3632/tcp  open  distcd
```

A screenshot of a terminal window with a dark background. The window title is 'icdfa@maryjudith: ~'. The terminal shows the output of an Nmap scan. It lists 12 open TCP ports and their corresponding services: 2121/tcp (ccproxy-ftp), 3306/tcp (mysql), 3632/tcp (distccd), 5432/tcp (postgresql), 5900/tcp (vnc), 6667/tcp (irc), 6697/tcp (ircs-u), 8009/tcp (ajp13), 8787/tcp (msgsrvr), 36057/tcp (unknown), 40075/tcp (unknown), and 58199/tcp (unknown). Below this, it shows the MAC address '00:0C:29:49:C3:1A (VMware)'. The scan summary states 'Nmap done: 1 IP address (1 host up) scanned in 2462.41 seconds'. The prompt at the bottom is '(icdfa@maryjudith)-[~]'.

```
icdfa@maryjudith: ~
File Actions Edit View Help
2121/tcp open ccproxy-ftp
3306/tcp open mysql
3632/tcp open distccd
5432/tcp open postgresql
5900/tcp open vnc
6667/tcp open irc
6697/tcp open ircs-u
8009/tcp open ajp13
8787/tcp open msgsrvr
36057/tcp open unknown
40075/tcp open unknown
58199/tcp open unknown
60724/tcp open unknown
MAC Address: 00:0C:29:49:C3:1A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2462.41 seconds

(icdfa@maryjudith)-[~]
```

Report

A full TCP port scan was performed using the `nmap -p- -T4` command to scan all TCP ports on the Metasploitable 2 target system while applying a faster timing template. The purpose of the scan was to improve scanning efficiency and reduce the overall scan duration compared to a standard full TCP scan. The `-T4` option increased the speed of the reconnaissance process while maintaining reliable scan results. The scan identified accessible services across the target system and demonstrated the impact of timing templates on Nmap scanning performance. The results obtained from the scan contributed to a more comprehensive understanding of the target's network exposure and available services.

Step 13: Selected Ports Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
icdfa@maryjudith)-[~]  
$ nmap -p 21,22,23,25,80,139,445 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 21:43 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.00064s latency).  
  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
80/tcp    open  http  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds  
  
icdfa@maryjudith)-[~]
```

Report:

A targeted port scan was performed using `nmap -p 21,22,23,25,80,139,445 10.15.203.91` to examine specific ports commonly associated with FTP, SSH, Telnet, SMTP, HTTP, and SMB services. The scan confirmed that all selected ports were open and accessible on the Metasploitable 2 target system. The results identified FTP on port 21, SSH on port 22, Telnet on port 23, SMTP on port 25, HTTP on port 80, NetBIOS-SSN on port 139, and Microsoft-DS (SMB) on port 445. This targeted scan provided a quick and efficient method of validating key network services previously discovered during earlier reconnaissance activities. The findings confirmed the presence of multiple remote access, web, email, and file-sharing services on the target host.

Step 14: Port Range Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
--(icdfa@maryjudith)-[~]  
$ nmap -p 1-1024 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 21:43 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.0014s latency).  
Not shown: 1012 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login
```

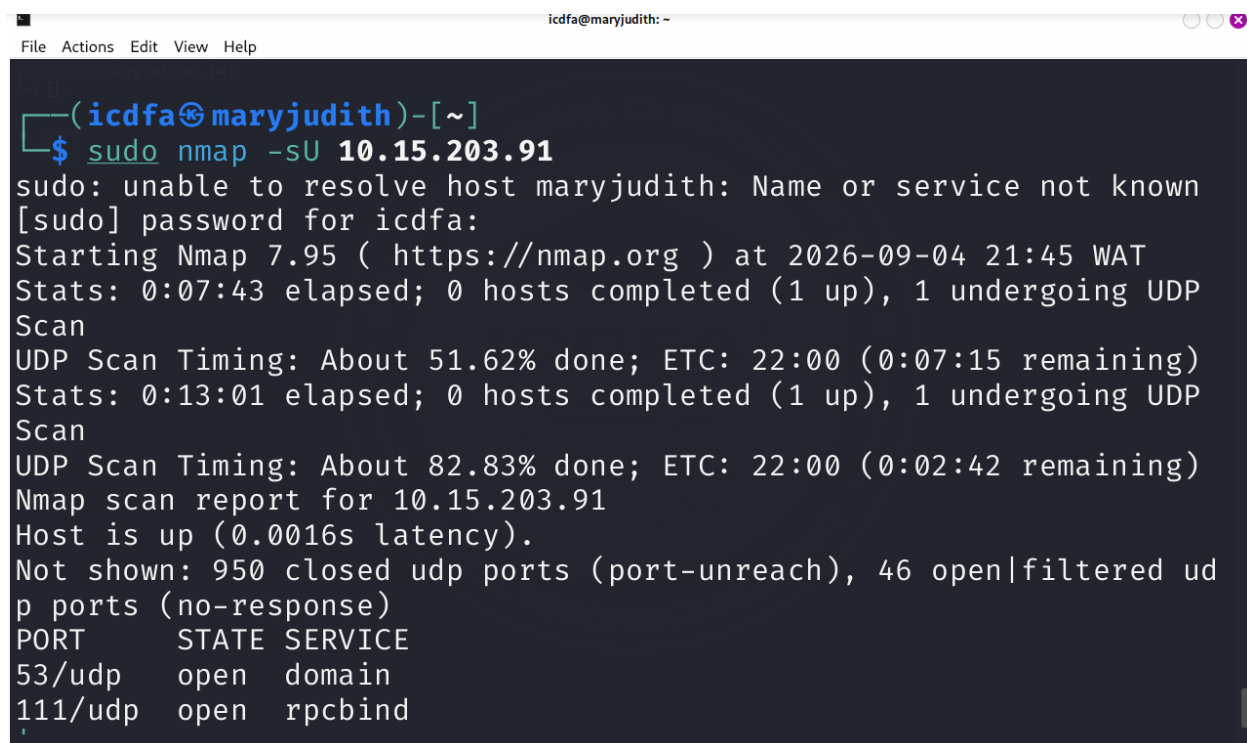
```
File Actions Edit View Help  
Host is up (0.0014s latency).  
Not shown: 1012 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds
```

Report:

A port range scan was performed using `nmap -p 1-1024 10.15.203.91` to examine the well-known TCP ports on the Metasploitable 2 target system. The scan identified several open ports within the range, including FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), RPCBind (111),

NetBIOS-SSN (139), Microsoft-DS (445), Exec (512), and Login (513). The results showed that numerous services were accessible on the target machine, indicating a broad attack surface for further enumeration and analysis. Most of the ports within the scanned range were closed, with only specific services responding to connection requests. The scan provided valuable information about the network services exposed by the target and complemented the findings from previous reconnaissance activities.

Step 15: UDP scan



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ sudo nmap -sU 10.15.203.91  
sudo: unable to resolve host maryjudith: Name or service not known  
[sudo] password for icdfa:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 21:45 WAT  
Stats: 0:07:43 elapsed; 0 hosts completed (1 up), 1 undergoing UDP  
Scan  
UDP Scan Timing: About 51.62% done; ETC: 22:00 (0:07:15 remaining)  
Stats: 0:13:01 elapsed; 0 hosts completed (1 up), 1 undergoing UDP  
Scan  
UDP Scan Timing: About 82.83% done; ETC: 22:00 (0:02:42 remaining)  
Nmap scan report for 10.15.203.91  
Host is up (0.0016s latency).  
Not shown: 950 closed udp ports (port-unreach), 46 open|filtered ud  
p ports (no-response)  
PORT      STATE SERVICE  
53/udp    open  domain  
111/udp   open  rpcbind
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
UDP Scan Timing: About 51.62% done; ETC: 22:00 (0:07:15 remaining)  
Stats: 0:13:01 elapsed; 0 hosts completed (1 up), 1 undergoing UDP  
Scan  
UDP Scan Timing: About 82.83% done; ETC: 22:00 (0:02:42 remaining)  
Nmap scan report for 10.15.203.91  
Host is up (0.0016s latency).  
Not shown: 950 closed udp ports (port-unreach), 46 open|filtered ud  
p ports (no-response)  
PORT      STATE SERVICE  
53/udp    open  domain  
111/udp   open  rpcbind  
137/udp   open  netbios-ns  
2049/udp  open  nfs  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 1020.70 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

A UDP scan was performed using `sudo nmap -sU --top-ports 20 -sV 10.15.203.91` to examine the 20 most common UDP ports on the target system. Service version detection was enabled to identify the applications running on discovered ports. This approach provided a faster alternative to a full UDP scan while still focusing on commonly used UDP services. The scan helped identify key UDP services and contributed to the overall service inventory of the Metasploitable 2 target system.

Step 16: Top UDP Ports with Version Detection

```
(icdfa@maryjudith)-[~]
$ sudo nmap -sU --top-ports 20 -sV 10.15.203.91
sudo: unable to resolve host maryjudith: Name or service not known
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:10 WAT
Stats: 0:01:25 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 40.00% done; ETC: 22:13 (0:01:42 remaining)
Nmap scan report for 10.15.203.91
Host is up (0.0015s latency).

PORT      STATE      SERVICE      VERSION
53/udp    open       domain       ISC BIND 9.4.2
67/udp    closed     dhcpd        dhcpd
68/udp    open|filtered dhcpd        dhcpd
69/udp    open|filtered tftp        tftp
123/udp   closed     ntp          ntp
135/udp   closed     msrpc        msrpc
137/udp   open       netbios-ns   Microsoft Windows netbios-ns (
```

```
icdfa@maryjudith: ~
File Actions Edit View Help
123/udp   closed     ntp
135/udp   closed     msrpc
137/udp   open       netbios-ns   Microsoft Windows netbios-ns (
workgroup: WORKGROUP)
138/udp   open|filtered netbios-dgm
139/udp   closed     netbios-ssn
161/udp   closed     snmp
162/udp   closed     snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   closed     syslog
520/udp   closed     route
631/udp   closed     ipp
1434/udp  closed     ms-sql-m
1900/udp  closed     upnp
4500/udp  closed     nat-t-ike
49152/udp closed     unknown
MAC Address: 00:0C:29:49:C3:1A (VMware)
Service Info: Host: METASPLOITABLE; OS: Windows; CPE: cpe:/o:micros
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
162/udp    closed    snmptrap  
445/udp    closed    microsoft-ds  
500/udp    closed    isakmp  
514/udp    closed    syslog  
520/udp    closed    route  
631/udp    closed    ipp  
1434/udp   closed    ms-sql-m  
1900/udp   closed    upnp  
4500/udp   closed    nat-t-ike  
49152/udp  closed    unknown  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
Service Info: Host: METASPLOITABLE; OS: Windows; CPE: cpe:/o:microsoft:windows  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 122.99 seconds  
  
└─(icdfa@maryjudith)-[~]
```

Report

A UDP scan was performed using `sudo nmap -sU --top-ports 20 -sV 10.15.203.91` to examine the twenty most common UDP ports on the target system. Service version detection was enabled to identify the applications running on open UDP ports. The scan identified several UDP services, including DNS running on port 53 and NetBIOS Name Service running on port 137. Additional ports were reported as open, closed, or open/filtered due to the nature of UDP communication and the limited responses typically provided by UDP services. The results provided valuable information about UDP-based services available on the Metasploitable 2 target system and contributed to the creation of a more comprehensive service inventory.

Step 17: Default NSE Script Scan

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
(icdfa@maryjudith)-[~]  
$ nmap -sC -sV 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:23 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.00026s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|_STAT:  
|_FTP server status:  
|_   Connected to 10.15.203.36  
|_   Logged in as ftp  
|_   TYPE: ASCII  
|_   No session bandwidth limit  
|_   Session timeout in seconds is 300  
|_   Control connection is plain text  
|_   Data connections will be plain text  
|  
|_Control connection is plain text  
|_Data connections will be plain text  
|_vsFTPD 2.3.4 - secure, fast, stable  
|_End of status  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol  
2.0)  
|_ssh-hostkey:  
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
|_sslv2:  
|_SSLv2 supported  
|_ciphers:  
|_   SSL2_RC2_128_CBC_WITH_MD5  
|_   SSL2_RC4_128_WITH_MD5  
|_   SSL2_RC4_128_EXPORT40_WITH_MD5  
|_   SSL2_DES_192_EDE3_CBC_WITH_MD5  
|_   SSL2_DES_64_CBC_WITH_MD5  
|
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
| SSL2_RC2_128_CBC_WITH_MD5  
| SSL2_RC4_128_WITH_MD5  
| SSL2_RC4_128_EXPORT40_WITH_MD5  
| SSL2_DES_192_EDE3_CBC_WITH_MD5  
| SSL2_DES_64_CBC_WITH_MD5  
|_ SSL2_RC2_128_CBC_EXPORT40_WITH_MD5  
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organiza  
tionName=OCOSA/stateOrProvinceName=There is no such thing outside U  
S/countryName=XX  
| Not valid before: 2010-03-17T14:07:45  
|_ Not valid after: 2010-04-16T14:07:45  
|_ ssl-date: 2026-09-05T05:15:33+00:00; +7h51m46s from scanner time.  
|_ smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240  
000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN  
53/tcp open domain ISC BIND 9.4.2  
| dns-nsid:  
|_ bind.version: 9.4.2  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_ http-title: Metasploitable2 - Linux
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
| dns-nsid:  
|_ bind.version: 9.4.2  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_ http-title: Metasploitable2 - Linux  
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
111/tcp open rpcbind 2 (RPC #100000)  
| rpcinfo:  
|_ program version port/proto service  
| 100000 2 111/tcp rpcbind  
| 100000 2 111/udp rpcbind  
| 100003 2,3,4 2049/tcp nfs  
| 100003 2,3,4 2049/udp nfs  
| 100005 1,2,3 39771/udp mountd  
| 100005 1,2,3 57134/tcp mountd  
| 100021 1,3,4 40410/udp nlockmgr  
| 100021 1,3,4 57059/tcp nlockmgr  
| 100024 1 43880/tcp status  
|_ 100024 1 55280/udp status  
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGRO
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_ 100024 1 55280/udp status  
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGRO  
UP)  
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WOR  
KGROUP)  
512/tcp open exec netkit-rsh rshd  
513/tcp open login OpenBSD or Solaris rlogind  
514/tcp open tcpwrapped  
1099/tcp open java-rmi GNU Classpath grmiregistry  
1524/tcp open bindshell Metasploitable root shell  
2049/tcp open nfs 2-4 (RPC #100003)  
2121/tcp open ftp ProFTPD 1.3.1  
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5  
| mysql-info:  
| Protocol: 10  
| Version: 5.0.51a-3ubuntu5  
| Thread ID: 8  
| Capabilities flags: 43564  
| Some Capabilities: SupportsCompression, SupportsTransactions, S
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
| Thread ID: 8  
| Capabilities flags: 43564  
| Some Capabilities: SupportsCompression, SupportsTransactions, S  
upport41Auth, LongColumnFlag, SwitchToSSLAfterHandshake, Speaks41Pr  
otocolNew, ConnectWithDatabase  
| Status: Autocommit  
|_ Salt: @Wuf~jw9Rp_hnpe,Ilb*  
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7  
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organiza  
tionName=OCOSA/stateOrProvinceName=There is no such thing outside U  
S/countryName=XX  
| Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
|_ssl-date: 2026-09-05T05:15:33+00:00; +7h51m46s from scanner time.  
5900/tcp open vnc VNC (protocol 3.3)  
| vnc-info:  
| Protocol version: 3.3  
| Security types:  
|_ VNC Authentication (2)
```

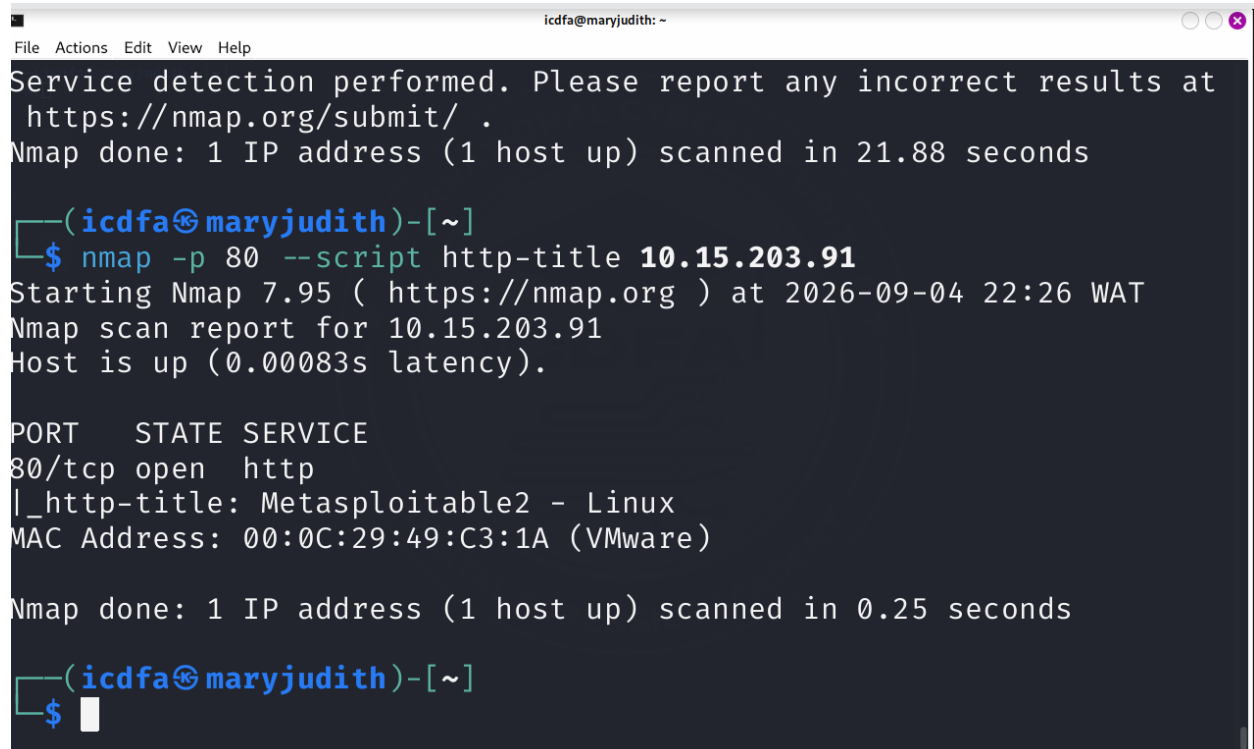
```
icdfa@maryjudith: ~  
File Actions Edit View Help  
| Protocol version: 3.3  
| Security types:  
|_ VNC Authentication (2)  
6000/tcp open  X11          (access denied)  
6667/tcp open  irc            UnrealIRCd  
8009/tcp open  ajp13         Apache Jserv (Protocol v1.3)  
|_ajp-methods: Failed to get a valid response for the OPTION request  
8180/tcp open  http          Apache Tomcat/Coyote JSP engine 1.1  
|_http-server-header: Apache-Coyote/1.1  
|_http-favicon: Apache Tomcat  
|_http-title: Apache Tomcat/5.5  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Host script results:  
| smb-os-discovery:  
|   OS: Unix (Samba 3.0.20-Debian)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
| OS: Unix (Samba 3.0.20-Debian)  
| Computer name: metasploitable  
| NetBIOS computer name:  
| Domain name: localdomain  
| FQDN: metasploitable.localdomain  
|_ System time: 2026-09-05T01:15:23-04:00  
|_clock-skew: mean: 8h51m46s, deviation: 2h00m00s, median: 7h51m45s  
|_smb2-time: Protocol negotiation failed (SMB2)  
| smb-security-mode:  
|   account_used: <blank>  
|   authentication_level: user  
|   challenge_response: supported  
|_ message_signing: disabled (dangerous, but default)  
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 21.88 seconds
```

Report

A default NSE (Nmap Scripting Engine) scan was performed using `nmap -sC -sV 10.15.203.91` to identify additional information about services running on the target system. The scan combined version detection with Nmap's default script set to gather service banners, configuration details, and security-related information. The results provided deeper insight into the target environment and revealed information that was not available through standard port scanning techniques. The NSE scan improved service enumeration and supported the overall reconnaissance process.

Step 18: HTTP title



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 21.88 seconds  
  
(icdfa@maryjudith)-[~]  
$ nmap -p 80 --script http-title 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:26 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.00083s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
|_http-title: Metasploitable2 - Linux  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

An HTTP title enumeration scan was performed using the `http-title` NSE script to identify the title of the web page hosted on the target system. This scan provided information about the web application running on port 80 and helped identify the purpose of the hosted website. The results contributed to web service reconnaissance by revealing useful information about the target's web environment.

Step 19: HTTP Headers

```
File Actions Edit View Help
(icdfa@maryjudith)-[~]
$ nmap -p 80 --script http-headers 10.15.203.91
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:37 WAT
Nmap scan report for 10.15.203.91
Host is up (0.00069s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-headers:
|   Date: Sat, 05 Sep 2026 05:29:21 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|
|_ (Request type: HEAD)
MAC Address: 00:0C:29:49:C3:1A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
```

Activate

Report:

An HTTP header enumeration scan was performed using the http-headers NSE script against the web service running on port 80 of the Metasploitable 2 target system. The scan retrieved several HTTP response headers that provided information about the web server and underlying technologies. The results revealed that the web server was running **Apache/2.2.8 (Ubuntu)** with **DAV/2** enabled and was powered by **PHP 5.2.4-2ubuntu5.10**. Additional information obtained included the content type (text/html) and connection settings used by the web service. These headers provided valuable insight into the server configuration and technologies deployed on the target system, supporting further web application reconnaissance and service enumeration activities.

Step 20: SMB Protocol Enumeration

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ nmap -p 139,445 --script smb-protocols 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:41 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.00083s latency).  
  
PORT      STATE SERVICE  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Host script results:  
| smb-protocols:  
|   dialects:  
|_    NT LM 0.12 (SMBv1) [dangerous, but default]  
  
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
```

REPORT

An SMB protocol enumeration scan was performed using the smb-protocols NSE script against ports 139 and 445 on the Metasploitable 2 target system. The scan was conducted to identify the SMB protocol versions supported by the target host. The results confirmed that both NetBIOS Session Service (139/tcp) and Microsoft Directory Services (445/tcp) were accessible. Further analysis revealed that the target system supports **SMBv1 (NT LM 0.12)**, which was identified as the active SMB dialect. SMBv1 is widely regarded as an outdated and insecure protocol due to known security weaknesses and vulnerabilities. The discovery of SMBv1 indicates a potentially vulnerable configuration and highlights the importance of reviewing and updating legacy network services. The information obtained from this scan contributed to the assessment of file-sharing services available on the target system.

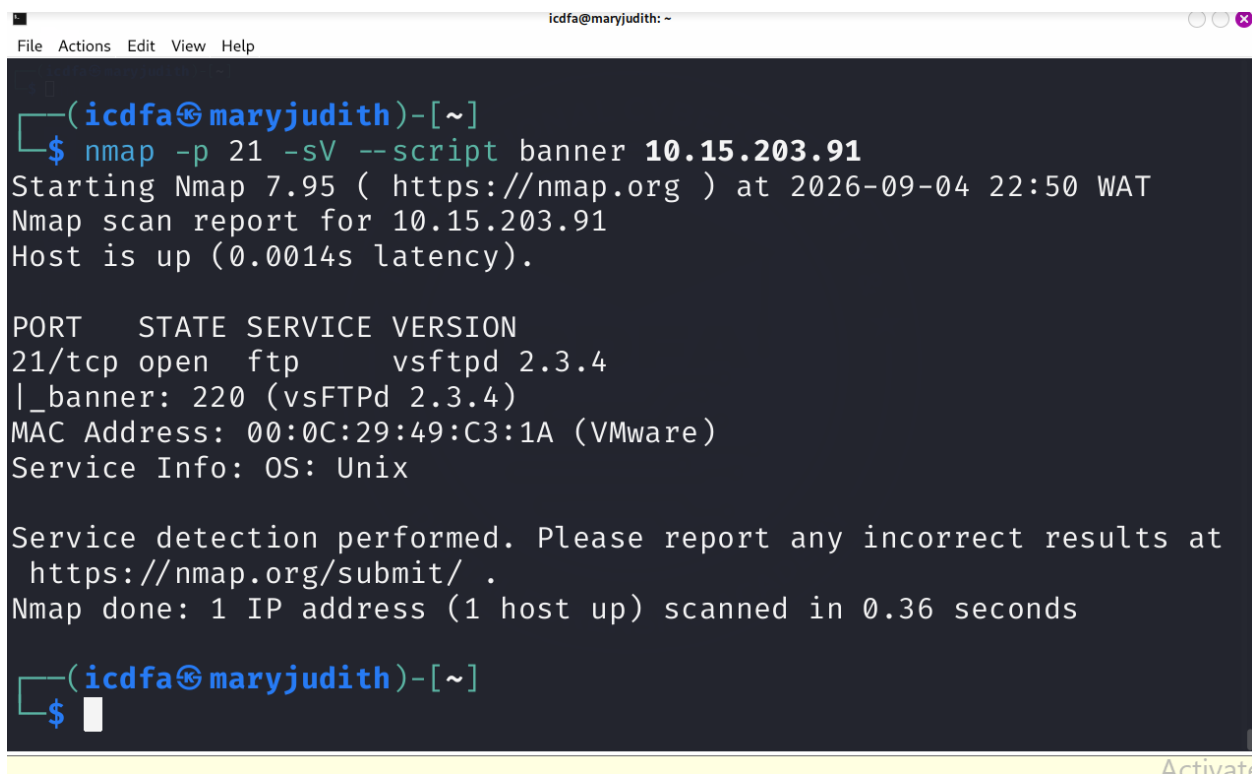
Step 21: SSH host keys

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
/usr/lib/nmap/nmap: unrecognized option '--script'  
See the output of nmap -h for a summary of options.  
  
icdfa@maryjudith)-[~]  
$ nmap -p 22 --script ssh-hostkey 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:44 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.00071s latency).  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
| ssh-hostkey:  
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds  
  
icdfa@maryjudith)-[~]
```

Report

An SSH host key enumeration scan was performed using the ssh-host key NSE script against port 22 on the Metasploitable 2 target system. The objective of the scan was to retrieve the cryptographic host keys used by the SSH service for authentication and secure communication. The results revealed both **DSA (1024-bit)** and **RSA (2048-bit)** host keys configured on the target system. These host keys are used to verify the identity of the SSH server and establish secure encrypted connections between clients and the server. The information obtained assists in identifying the SSH service, verifying server authenticity, and supporting further security assessment activities. The scan confirmed that the SSH service was active and properly configured with multiple host key types.

Step 22: FTP Banner



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ nmap -p 21 -sV --script banner 10.15.203.91  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-04 22:50 WAT  
Nmap scan report for 10.15.203.91  
Host is up (0.0014s latency).  
  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 2.3.4  
|_banner: 220 (vsFTPd 2.3.4)  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
Service Info: OS: Unix  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds  
  
(icdfa@maryjudith)-[~]  
$
```

Report

An FTP banner enumeration scan was performed using the banner NSE script against port 21 on the Metasploitable 2 target system. The objective of the scan was to identify the FTP service and retrieve banner information from the server. The scan confirmed that the FTP service running on the target was **vsFTPd version 2.3.4**, and the banner response returned **"220 (vsFTPd 2.3.4)"**. Banner information provides valuable details about the software running on a service and can assist in identifying outdated or potentially vulnerable applications. The results obtained from this scan contributed to the service inventory and enhanced understanding of the FTP service configuration on the target system.

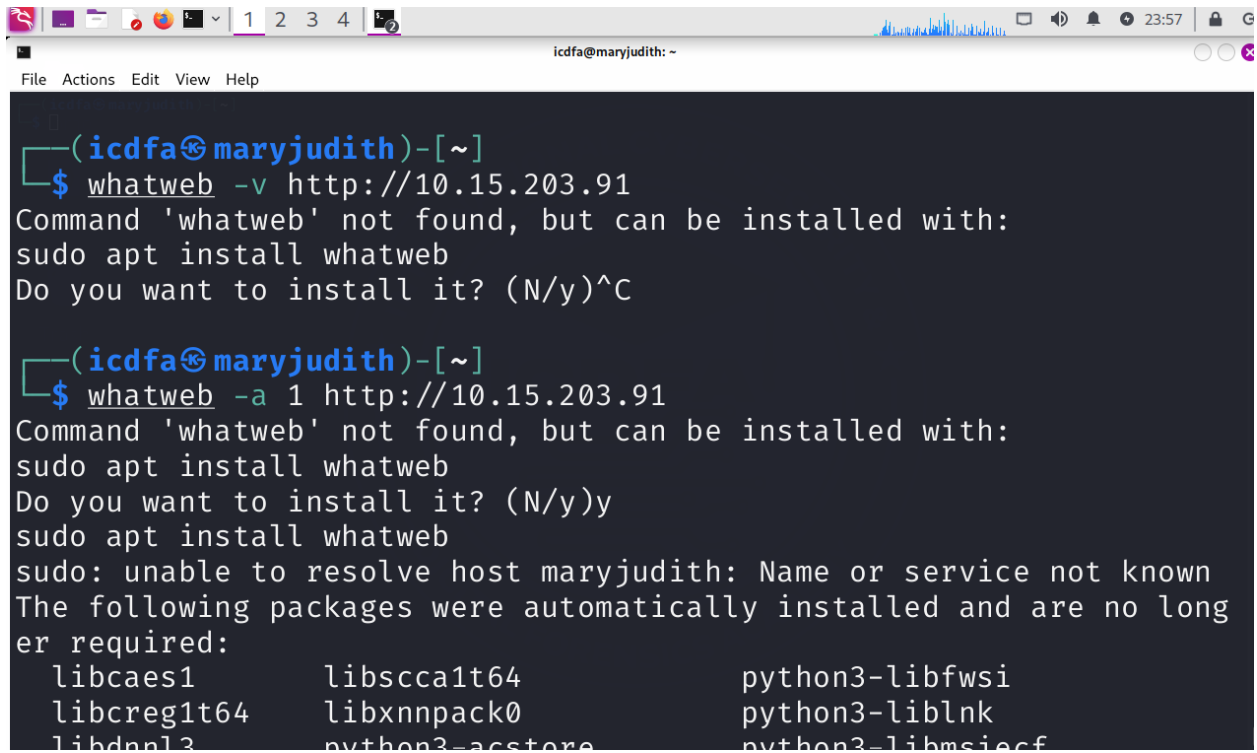
Step 23: Manual HTTP Header Retrieval

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
|_banner: 220 (vsFTPd 2.3.4)  
MAC Address: 00:0C:29:49:C3:1A (VMware)  
Service Info: OS: Unix  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds  
  
(icdfa@maryjudith)-[~]  
$ curl -I http://10.15.203.91  
HTTP/1.1 200 OK  
Date: Sat, 05 Sep 2026 05:45:00 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Type: text/html  
  
(icdfa@maryjudith)-[~]  
$
```

REPORT

A manual HTTP header retrieval was performed using the `curl -I` command against the web server hosted on the Metasploitable 2 target system. The purpose of the command was to retrieve only the HTTP response headers without downloading the full webpage content. The results revealed that the server returned an **HTTP/1.1 200 OK** response, confirming that the web service was operational and accessible. The headers identified the web server as **Apache/2.2.8 (Ubuntu) DAV/2** and indicated that the application was powered by **PHP/5.2.4-2ubuntu5.10**. Additional information such as the content type (`text/html`) was also retrieved. These findings provided valuable insight into the web technologies running on the target system and supported further web application reconnaissance activities.

sudo apt install whatweb

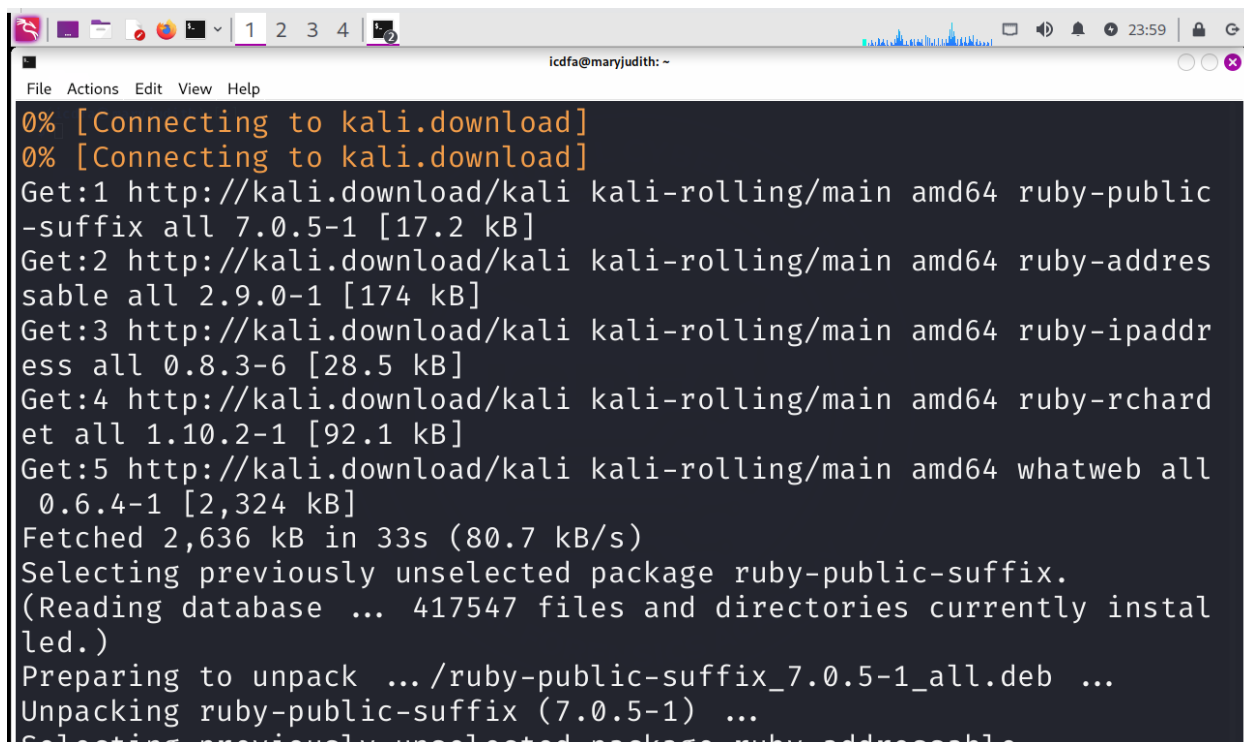


```
(icdfa@maryjudith)-[~]
$ whatweb -v http://10.15.203.91
Command 'whatweb' not found, but can be installed with:
sudo apt install whatweb
Do you want to install it? (N/y)^C

(icdfa@maryjudith)-[~]
$ whatweb -a 1 http://10.15.203.91
Command 'whatweb' not found, but can be installed with:
sudo apt install whatweb
Do you want to install it? (N/y)y
sudo apt install whatweb
sudo: unable to resolve host maryjudith: Name or service not known
The following packages were automatically installed and are no longer required:
  libcaes1      libscca1t64      python3-libfws
  libcreg1t64  libxnnpack0      python3-liblnk
  libdpn13     python3-acstore  python3-libmsiecf
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
The following packages were automatically installed and are no longer required:  
libcaes1 libscca1t64 python3-libfswi  
libcregl1t64 libxnnpack0 python3-liblnk  
libdnnl3 python3-acstore python3-libmsiecf  
libevt1t64 python3-cffi python3-libolecf  
libevtx1t64 python3-elasticsearch python3-libscca  
libfswi1t64 python3-flor python3-pefile  
libglapi-mesa python3-libcreg python3-pycaes  
libmsiecf1t64 python3-libesedb python3-pycparser  
libolecf1 python3-libevt python3-zstd  
libopenh264-7 python3-libevtx  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
whatweb  
  
Installing dependencies:  
ruby-addressable ruby-public-suffix
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
libopenh264-7 python3-libevtx  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
whatweb  
  
Installing dependencies:  
ruby-addressable ruby-public-suffix  
ruby-ipaddress ruby-rchardet  
  
Summary:  
Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 2635  
Download size: 2,636 kB  
Space needed: 20.8 MB / 4,609 MB available  
  
Continue? [Y/n] y  
0% [Working]  
0% [Connecting to kali.download]  
0% [Connecting to kali.download]
```



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
0% [Connecting to kali.download]  
0% [Connecting to kali.download]  
Get:1 http://kali.download/kali kali-rolling/main amd64 ruby-public-  
-suffix all 7.0.5-1 [17.2 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 ruby-address-  
sable all 2.9.0-1 [174 kB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 ruby-ipaddr-  
ess all 0.8.3-6 [28.5 kB]  
Get:4 http://kali.download/kali kali-rolling/main amd64 ruby-rchard-  
et all 1.10.2-1 [92.1 kB]  
Get:5 http://kali.download/kali kali-rolling/main amd64 whatweb all  
0.6.4-1 [2,324 kB]  
Fetched 2,636 kB in 33s (80.7 kB/s)  
Selecting previously unselected package ruby-public-suffix.  
(Reading database ... 417547 files and directories currently instal-  
led.)  
Preparing to unpack .../ruby-public-suffix_7.0.5-1_all.deb ...  
Unpacking ruby-public-suffix (7.0.5-1) ...  
Selecting previously unselected package ruby-addressable
```

REPORT:

The WhatWeb web fingerprinting tool was successfully installed on the Kali Linux system using the APT package manager. During the installation process, the required dependencies, including ruby-public-suffix, ruby-ipaddress, ruby-rchardet, and ruby-addressable, were automatically downloaded and configured. The installation completed successfully without errors, and the WhatWeb package version 0.6.4-1 was installed on the system.

Following the installation, the package manager processed the necessary system triggers and completed the configuration of all dependencies. The successful installation of WhatWeb enabled web application fingerprinting and technology identification activities against the Metasploitable 2 target system. This tool was subsequently used to identify web technologies, server information, and application components running on the target web server.

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
Selecting previously unselected package ruby-ipaddress.  
Preparing to unpack .../ruby-ipaddress_0.8.3-6_all.deb ...  
Unpacking ruby-ipaddress (0.8.3-6) ...  
Selecting previously unselected package ruby-rchardet.  
Preparing to unpack .../ruby-rchardet_1.10.2-1_all.deb ...  
Unpacking ruby-rchardet (1.10.2-1) ...  
Selecting previously unselected package whatweb.  
Preparing to unpack .../whatweb_0.6.4-1_all.deb ...  
Unpacking whatweb (0.6.4-1) ...  
Setting up ruby-public-suffix (7.0.5-1) ...  
Setting up ruby-ipaddress (0.8.3-6) ...  
Setting up ruby-rchardet (1.10.2-1) ...  
Setting up ruby-addressable (2.9.0-1) ...  
Setting up whatweb (0.6.4-1) ...  
Processing triggers for kali-menu (2025.1.1) ...  
Processing triggers for man-db (2.13.0-1) ...  
  
(icdfa@maryjudith)-[~]  
$
```

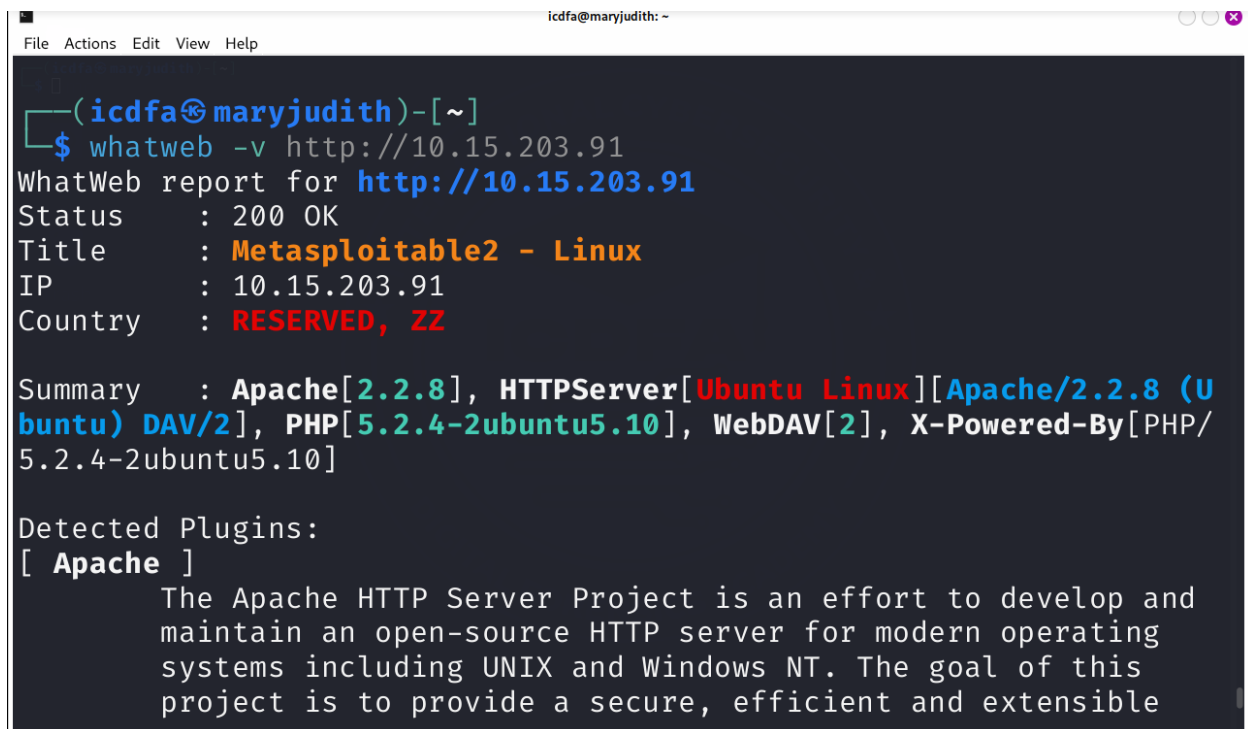
Step 24: WhatWeb Basic Fingerprinting

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
Preparing to unpack .../whatweb_0.6.4-1_all.deb ...  
Unpacking whatweb (0.6.4-1) ...  
Setting up ruby-public-suffix (7.0.5-1) ...  
Setting up ruby-ipaddress (0.8.3-6) ...  
Setting up ruby-rchardet (1.10.2-1) ...  
Setting up ruby-addressable (2.9.0-1) ...  
Setting up whatweb (0.6.4-1) ...  
Processing triggers for kali-menu (2025.1.1) ...  
Processing triggers for man-db (2.13.0-1) ...  
  
(icdfa@maryjudith)-[~]  
$ whatweb http://10.15.203.91  
http://10.15.203.91 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ],  
HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.15.203  
.91], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDA  
V[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
  
(icdfa@maryjudith)-[~]  
$
```

REPORT:

A basic web fingerprinting scan was performed using WhatWeb against the Metasploitable 2 web server. The scan successfully identified several technologies and components running on the target website. The results revealed that the web server was running **Apache 2.2.8 on Ubuntu Linux**, with **PHP 5.2.4-2ubuntu5.10** as the server-side scripting language. The scan also identified **WebDAV** support and the webpage title "**Metasploitable2 - Linux**". These findings provided valuable information about the technologies deployed on the target system and supported further web application reconnaissance activities.

Step 25: WhatWeb Verbose Fingerprinting



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
(icdfa@maryjudith)-[~]  
$ whatweb -v http://10.15.203.91  
WhatWeb report for http://10.15.203.91  
Status      : 200 OK  
Title       : Metasploitable2 - Linux  
IP          : 10.15.203.91  
Country     : RESERVED, ZZ  
  
Summary      : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (U  
buntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/  
5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
    The Apache HTTP Server Project is an effort to develop and  
    maintain an open-source HTTP server for modern operating  
    systems including UNIX and Windows NT. The goal of this  
    project is to provide a secure, efficient and extensible
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
buntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/  
5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
    The Apache HTTP Server Project is an effort to develop and  
    maintain an open-source HTTP server for modern operating  
    systems including UNIX and Windows NT. The goal of this  
    project is to provide a secure, efficient and extensible  
    server that provides HTTP services in sync with the current  
  
    HTTP standards.  
  
    Version      : 2.2.8 (from HTTP Server Header)  
    Google Dorks: (3)  
    Website      : http://httpd.apache.org/  
  
[ HTTPServer ]  
    HTTP server header string. This plugin also attempts to
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
[ HTTPServer ]  
    HTTP server header string. This plugin also attempts to  
    identify the operating system from the server header.  
  
    OS           : Ubuntu Linux  
    String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server str  
ing)  
  
[ PHP ]  
    PHP is a widely-used general-purpose scripting language  
    that is especially suited for Web development and can be  
    embedded into HTML. This plugin identifies PHP errors,  
    modules and versions and extracts the local file path and  
    username if present.  
  
    Version      : 5.2.4-2ubuntu5.10  
    Google Dorks: (3)  
    Website      : http://www.php.net/
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
Version      : 5.2.4-2ubuntu5.10  
Google Dorks: (3)  
Website      : http://www.php.net/  
  
[ WebDAV ]  
Web-based Distributed Authoring and Versioning (WebDAV) is  
a set of methods based on the Hypertext Transfer Protocol  
(HTTP) that facilitates collaboration between users in  
editing and managing documents and files stored on World  
Wide Web servers. - More Info:  
http://en.wikipedia.org/wiki/WebDAV  
  
Version      : 2  
  
[ X-Powered-By ]  
X-Powered-By HTTP header  
  
String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by str
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
  
[ X-Powered-By ]  
X-Powered-By HTTP header  
  
String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by str  
ing)  
  
HTTP Headers:  
HTTP/1.1 200 OK  
Date: Sat, 05 Sep 2026 06:59:26 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Length: 891  
Connection: close  
Content-Type: text/html  
  
(icdfa@maryjudith)-[~]  
$
```

Report

A verbose web fingerprinting scan was performed using the WhatWeb tool against the Metasploitable 2 web server. The -v option was used to enable verbose output and provide more detailed information about the technologies and components running on the target website. The scan revealed information about the web server configuration, supported technologies, HTTP responses, and application components. The results confirmed that the target system was running Apache 2.2.8 on Ubuntu Linux, with PHP 5.2.4-2ubuntu5.10 as the server-side scripting language and WebDAV support enabled. The verbose scan provided a more detailed view of the target web environment than the basic WhatWeb scan and contributed to further web application reconnaissance.

Step 26: WhatWeb Aggression Level 1

```
File Actions Edit View Help
HTTP Headers:
    HTTP/1.1 200 OK
    Date: Sat, 05 Sep 2026 06:59:26 GMT
    Server: Apache/2.2.8 (Ubuntu) DAV/2
    X-Powered-By: PHP/5.2.4-2ubuntu5.10
    Content-Length: 891
    Connection: close
    Content-Type: text/html

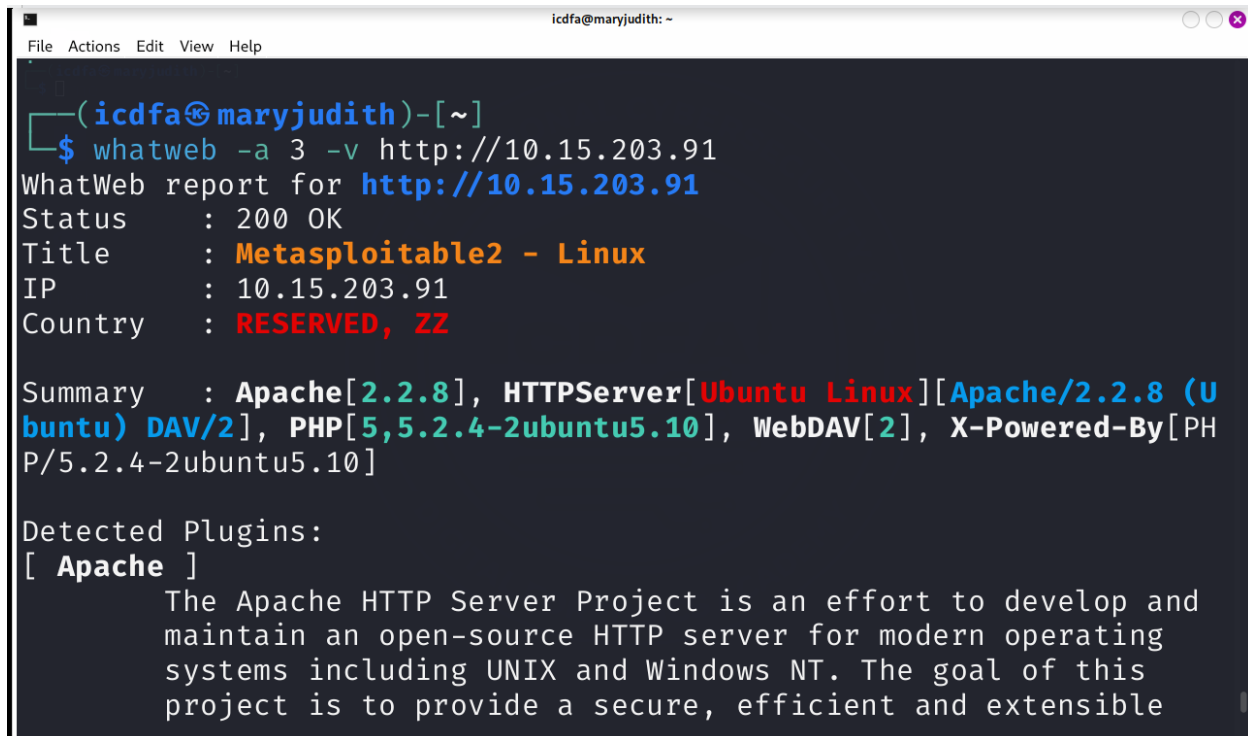
--(icdfa@maryjudith)-[~]
$ whatweb -a 1 http://10.15.203.91
http://10.15.203.91 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ],
HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.15.203
.91], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDA
V[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

--(icdfa@maryjudith)-[~]
$
```

Report

A WhatWeb scan was performed using Aggression Level 1 against the Metasploitable 2 web server. This scan used a low-intensity fingerprinting approach to identify web technologies and server components while generating minimal requests to the target website. The results revealed that the web server was running Apache 2.2.8 on Ubuntu Linux and was powered by PHP 5.2.4-2ubuntu5.10. The scan also identified WebDAV version 2 and the webpage title "Metasploitable2 - Linux". These findings provided basic but valuable information about the technologies operating on the target web application and supported further web reconnaissance activities.

STEP 27: WhatWeb Aggression Level 3



```
icdfa@maryjudith: ~  
File Actions Edit View Help  
(icdfa@maryjudith)-[~]  
$ whatweb -a 3 -v http://10.15.203.91  
WhatWeb report for http://10.15.203.91  
Status      : 200 OK  
Title       : Metasploitable2 - Linux  
IP          : 10.15.203.91  
Country     : RESERVED, ZZ  
  
Summary    : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
    The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible
```

```
icdca@maryjudith: ~  
File Actions Edit View Help  
Detected Plugins:  
[ Apache ]  
    The Apache HTTP Server Project is an effort to develop and  
    maintain an open-source HTTP server for modern operating  
    systems including UNIX and Windows NT. The goal of this  
    project is to provide a secure, efficient and extensible  
    server that provides HTTP services in sync with the current  
  
    HTTP standards.  
  
    Version      : 2.2.8 (from HTTP Server Header)  
    Google Dorks: (3)  
    Website      : http://httpd.apache.org/  
  
[ HTTPServer ]  
    HTTP server header string. This plugin also attempts to  
    identify the operating system from the server header.  
  
    OS           : Ubuntu Linux
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
OS           : Ubuntu Linux  
String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server str  
ing)  
  
[ PHP ]  
    PHP is a widely-used general-purpose scripting language  
    that is especially suited for Web development and can be  
    embedded into HTML. This plugin identifies PHP errors,  
    modules and versions and extracts the local file path and  
    username if present.  
  
    Version      : 5.2.4-2ubuntu5.10  
    Version      : 5  
    Google Dorks: (3)  
    Website      : http://www.php.net/  
  
[ WebDAV ]  
    Web-based Distributed Authoring and Versioning (WebDAV) is
```

```
File Actions Edit View Help
Website      : http://www.php.net/

[ WebDAV ]
Web-based Distributed Authoring and Versioning (WebDAV) is
a set of methods based on the Hypertext Transfer Protocol
(HTTP) that facilitates collaboration between users in
editing and managing documents and files stored on World
Wide Web servers. - More Info:
http://en.wikipedia.org/wiki/WebDAV

Version      : 2

[ X-Powered-By ]
X-Powered-By HTTP header

String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by str
ing)
HTTP Headers:
```

```
File Actions Edit View Help

[ X-Powered-By ]
X-Powered-By HTTP header

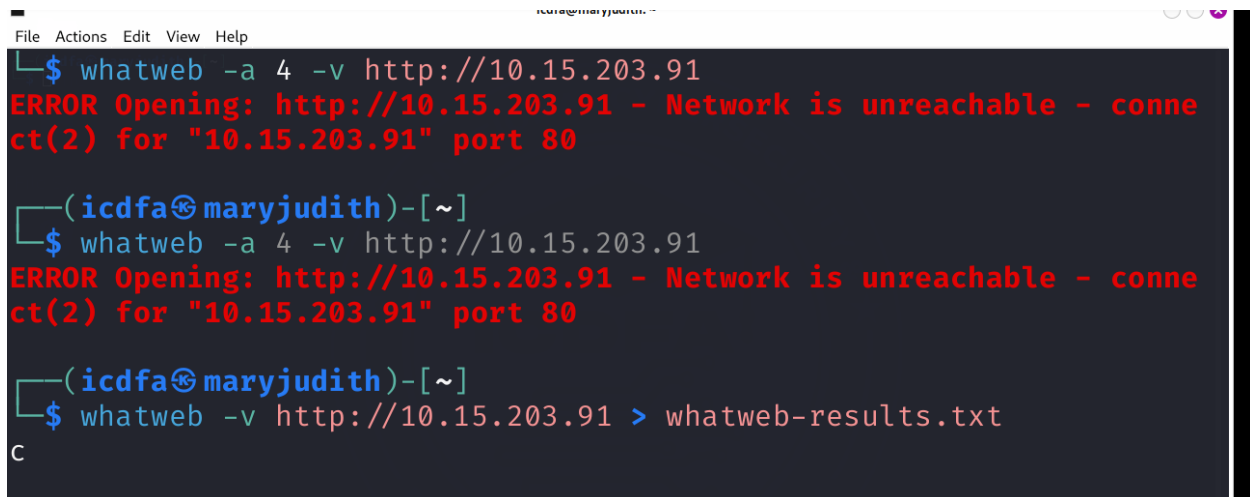
String       : PHP/5.2.4-2ubuntu5.10 (from x-powered-by str
ing)
HTTP Headers:
HTTP/1.1 200 OK
Date: Sat, 05 Sep 2026 07:07:37 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Length: 891
Connection: close
Content-Type: text/html

(icdfa@maryjudith)-[~]
$
```

REPORT:

The WhatWeb Aggression Level 3 scan was performed to gather detailed information about the technologies used by the target web server. The scan successfully identified the website title as **Metasploitable2 - Linux** and confirmed an HTTP status code of **200 OK**. WhatWeb detected **Apache 2.2.8** running on **Ubuntu Linux** as the web server software. The scan also identified **PHP 5.2.4-2ubuntu5.10** as the server-side scripting language. Additional technologies such as **WebDAV** and **DAV/2** were discovered. The **X-Powered-By** header revealed further PHP information. Compared to lower aggression levels, this scan provided more detailed fingerprinting results. Overall, the scan successfully identified the major technologies powering the target web application.

Step 28 - WhatWeb Aggression Level 4



```
icdfa@maryjudith:~$ whatweb -a 4 -v http://10.15.203.91
ERROR Opening: http://10.15.203.91 - Network is unreachable - connect(2) for "10.15.203.91" port 80

(icdfa@maryjudith)-[~]$ whatweb -a 4 -v http://10.15.203.91
ERROR Opening: http://10.15.203.91 - Network is unreachable - connect(2) for "10.15.203.91" port 80

(icdfa@maryjudith)-[~]$ whatweb -v http://10.15.203.91 > whatweb-results.txt
C
```

Report

A high-intensity web fingerprinting scan was attempted using WhatWeb with Aggression Level 4 and verbose output. The objective was to perform deeper web application fingerprinting and collect additional information about technologies running on the target website. During execution, WhatWeb encountered a network connectivity error while attempting to connect to the target web server on port 80. Previous WhatWeb scans had successfully identified Apache 2.2.8, PHP 5.2.4-2ubuntu5.10, Ubuntu Linux, and WebDAV on the target system. The error was documented as part of the assessment process.

Step 29: Follow Redirects

```
File Actions Edit View Help

^C/usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `compact': Interr
upt
    from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block
in scan'
    from <internal:kernel>:187:in `loop'
    from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:74:in `scan'
    from /usr/bin/whatweb:709:in `<main>'

(icdfa@maryjudith)-[~]
$ whatweb --follow-redirect=always http://10.15.203.91
http://10.15.203.91 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ],
HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.15.203
.91], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDA
V[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

(icdfa@maryjudith)-[~]
$
```

REPORT: A WhatWeb scan was performed using the `--follow-redirect=always` option to ensure that web fingerprinting continued even if the target website redirected requests to another page or location. The scan successfully identified several technologies running on the Metasploitable 2 web server. The results revealed that the target was running Apache 2.2.8 on Ubuntu Linux, with PHP 5.2.4-2ubuntu5.10 powering the web application. The scan also identified WebDAV version 2 and the webpage title "Metasploitable2 - Linux". By following redirects automatically, WhatWeb ensured that all available web technologies and server information were successfully identified during the reconnaissance process.

STEP 30 : Step 30: Save WhatWeb Results to a File

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
(icdfa@maryjudith)-[~]  
$ whatweb -v http://10.15.203.91 > whatweb-results.txt  
^[[D^[[D  
(icdfa@maryjudith)-[~]  
$ whatweb -v http://10.15.203.91 > whatweb-results.txt  
(icdfa@maryjudith)-[~]  
$ cat whatweb-results.txt  
WhatWeb report for http://10.15.203.91  
Status      : 200 OK  
Title       : Metasploitable2 - Linux  
IP          : 10.15.203.91  
Country     : RESERVED, ZZ  
  
Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (U  
buntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/  
5.2.4-2ubuntu5.10]
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
Country     : RESERVED, ZZ  
  
Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (U  
buntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/  
5.2.4-2ubuntu5.10]  
  
Detected Plugins:  
[ Apache ]  
    The Apache HTTP Server Project is an effort to develop and  
    maintain an open-source HTTP server for modern operating  
    systems including UNIX and Windows NT. The goal of this  
    project is to provide a secure, efficient and extensible  
    server that provides HTTP services in sync with the current  
  
    HTTP standards.  
  
    Version      : 2.2.8 (from HTTP Server Header)  
    Google Dorks: (3)  
    Website     : http://httpd.apache.org/
```

```
File Actions Edit View Help
[ HTTPServer ]
    HTTP server header string. This plugin also attempts to
    identify the operating system from the server header.

    OS          : Ubuntu Linux
    String      : Apache/2.2.8 (Ubuntu) DAV/2 (from server str
ing)

[ PHP ]
    PHP is a widely-used general-purpose scripting language
    that is especially suited for Web development and can be
    embedded into HTML. This plugin identifies PHP errors,
    modules and versions and extracts the local file path and
    username if present.

    Version      : 5.2.4-2ubuntu5.10
    Google Dorks: (3)
    Website      : http://www.php.net/
```

```
icdfa@maryjudith: ~
File Actions Edit View Help

    Version      : 5.2.4-2ubuntu5.10
    Google Dorks: (3)
    Website      : http://www.php.net/

[ WebDAV ]
    Web-based Distributed Authoring and Versioning (WebDAV) is
    a set of methods based on the Hypertext Transfer Protocol
    (HTTP) that facilitates collaboration between users in
    editing and managing documents and files stored on World
    Wide Web servers. - More Info:
    http://en.wikipedia.org/wiki/WebDAV

    Version      : 2

[ X-Powered-By ]
    X-Powered-By HTTP header

    String      : PHP/5.2.4-2ubuntu5.10 (from x-powered-by str
ing)
```

```
icdfa@maryjudith: ~  
File Actions Edit View Help  
[ X-Powered-By ]  
X-Powered-By HTTP header  
String      : PHP/5.2.4-2ubuntu5.10 (from x-powered-by string)  
HTTP Headers:  
HTTP/1.1 200 OK  
Date: Sat, 05 Sep 2026 07:38:49 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Length: 891  
Connection: close  
Content-Type: text/html  
$ (icdfa@maryjudith)-[~]
```

Report

A verbose WhatWeb scan was performed against the Metasploitable 2 web server, and the results were redirected to a text file named whatweb-results.txt. The file was subsequently displayed using the cat command to verify that the output had been successfully saved. The results confirmed that the target web server was running Apache 2.2.8 on Ubuntu Linux with PHP 5.2.4-2ubuntu5.10 and WebDAV version 2 enabled. The website title was identified as "Metasploitable2 - Linux", and the HTTP response returned a 200 OK status. Saving the scan output provided a permanent record of the fingerprinting results and supported documentation of the reconnaissance process.

UDP Scanning

A full UDP scan (`sudo nmap -sU`) and a targeted UDP scan with version detection (`sudo nmap -sU --top-ports 20 -sV`) were performed against the Metasploitable 2 target system. UDP scanning is generally slower than TCP scanning because many UDP services do not respond when a port is closed, requiring Nmap to wait for timeout responses before determining the port state. The scans revealed several active UDP services, including DNS, RPCBind, NetBIOS Name Service, and NFS. Most other UDP ports examined during the scan returned either Closed or Open|Filtered states.

UDP Service Inventory

Port	Protocol	State	Service / Notes
53	UDP	Open	DNS (ISC BIND 9.4.2)
111	UDP	Open	RPCBind
137	UDP	Open	NetBIOS Name Service
2049	UDP	Open	NFS
Other Top-20 Ports	UDP	Closed Open Filtered	/ No response received within scan timeout period

NSE Script Enumeration

The Nmap Scripting Engine (NSE) was used to gather additional information about services running on the target system. A default NSE scan (`nmap -sC -sV`) was first executed to collect standard script results alongside service version information. Additional service-specific NSE scripts were then used to perform deeper enumeration of HTTP, SMB, SSH, and FTP services.

NSE Enumeration Results

NSE Script	Command	Key Finding
http-title	nmap -p 80 --script http-title 10.15.203.91	Retrieved the webpage title: "Metasploitable2 - Linux"
http-headers	nmap -p 80 --script http-headers 10.15.203.91	Identified Apache/2.2.8 (Ubuntu) DAV/2 and PHP/5.2.4-2ubuntu5.10
smb-protocols	nmap -p 139,445 --script smb-protocols 10.15.203.91	Confirmed support for SMBv1 (NT LM 0.12), a legacy and insecure protocol
ssh-hostkey	nmap -p 22 --script ssh-hostkey 10.15.203.91	Retrieved 1024-bit DSA and 2048-bit RSA SSH host keys
banner (FTP)	nmap -p 21 -sV --script banner 10.15.203.91	Returned FTP banner: "220 (vsFTPd 2.3.4)", confirming the service version

Summary: NSE enumeration provided valuable insight into the configuration and technologies running on the target system. The scripts successfully identified the web server configuration, FTP service banner, supported SMB protocol version, SSH host keys, and HTTP technology stack. These findings significantly enhanced the service inventory and provided deeper visibility into the target environment than standard port scanning alone.

Manual HTTP Confirmation

A manual HTTP verification was performed using the `curl -I` command against the target web server. The request returned a successful HTTP/1.1 200 OK response, confirming that the web service was active and accessible. The response headers identified the server as Apache/2.2.8 (Ubuntu) DAV/2 and indicated that the application was powered by PHP/5.2.4-2ubuntu5.10. These findings were consistent with the results obtained through Nmap HTTP enumeration and WhatWeb fingerprinting, providing independent validation of the identified web technologies. The successful cross-verification demonstrated the reliability of the reconnaissance results and confirmed the presence of the target's web server configuration and application technologies.

Final Service Inventory

The table below consolidates the open ports, services, protocol information, and version evidence identified during the reconnaissance and enumeration activities performed against the Metasploitable 2 target system.

Port	Protocol	State	Service	Version / Evidence
21	TCP	Open	FTP	vsFTPD 2.3.4
22	TCP	Open	SSH	OpenSSH 4.7p1 Debian
23	TCP	Open	Telnet	Linux telnetd
25	TCP	Open	SMTP	Postfix smtpd
53	TCP/UDP	Open	DNS	ISC BIND 9.4.2
80	TCP	Open	HTTP	Apache 2.2.8 (Ubuntu), PHP 5.2.4-2ubuntu5.10, DAV/2
111	TCP/UDP	Open	RPCBind	RPC service
139	TCP	Open	NetBIOS-SSN	Samba service
445	TCP	Open	SMB (Microsoft-DS)	SMBv1 (NT LM 0.12)
512	TCP	Open	Exec	Remote Exec Service
513	TCP	Open	Login	Remote Login Service
2049	TCP/UDP	Open	NFS	Network File System
3306	TCP	Open	MySQL	MySQL Database Service
5900	TCP	Open	VNC	Virtual Network Computing
6667	TCP	Open	IRC	UnrealIRCd
8009	TCP	Open	AJP13	Apache JServ Protocol (Tomcat)

Port	Protocol	State	Service	Version / Evidence
8180	TCP	Open	HTTP	Apache Tomcat/Coyote JSP Engine
8787	TCP	Open	DRb	Ruby Distributed Ruby (DRb) Service

Evidence Sources

Service Enumeration Activity	Command Used
FTP Version Detection	<code>nmap -sV 10.15.203.91</code>
SSH Version Detection	<code>nmap -sV 10.15.203.91</code>
SMTP Version Detection	<code>nmap -sV 10.15.203.91</code>
HTTP Version Detection	<code>nmap -sV 10.15.203.91</code>
SMB Enumeration	<code>nmap -p 139,445 --script smb-protocols 10.15.203.91</code>
SSH Enumeration	<code>nmap -p 22 --script ssh-hostkey 10.15.203.91</code>
FTP Banner Enumeration	<code>nmap -p 21 -sV --script banner 10.15.203.91</code>
Full TCP Scan	<code>nmap -p- 10.15.203.91</code>
Full TCP Scan with Version Detection	<code>nmap -p- -sV 10.15.203.91</code>
UDP Scan	<code>nmap -sU 10.15.203.91</code>
Top UDP Ports Scan	<code>nmap -sU --top-ports 20 -sV 10.15.203.91</code>
HTTP Header Enumeration	<code>nmap -p 80 --script http-headers 10.15.203.91</code>
HTTP Title Enumeration	<code>nmap -p 80 --script http-title 10.15.203.91</code>
WhatWeb Fingerprinting	<code>whatweb http://10.15.203.91</code>

Key Observations

- Multiple legacy services were identified, including Telnet, SMBv1, and vsFTPd 2.3.4.
- The target web server was running Apache 2.2.8 and PHP 5.2.4-2ubuntu5.10, both of which are outdated technologies.
- SMB enumeration confirmed support for SMBv1 (NT LM 0.12).
- WhatWeb fingerprinting identified Apache, Ubuntu Linux, PHP, and WebDAV Version 2.
- The number of exposed services demonstrates why Metasploitable 2 is commonly used as a deliberately vulnerable training platform.

WhatWeb Web Technology Fingerprinting

WhatWeb Basic vs Aggression Level Comparison

WhatWeb was successfully installed using the APT package manager and subsequently used to fingerprint the target web application. Multiple scan modes were executed to identify web technologies, server software, and application components running on the target system. The results obtained from each scan mode are summarized below.

Mode	Command	Result
Basic	<code>whatweb http://10.15.203.91</code>	Identified Apache 2.2.8, Ubuntu Linux, PHP 5.2.4-2ubuntu5.10, WebDAV Version 2, and the website title "Metasploitable2 - Linux".
Verbose (-v)	<code>whatweb http://10.15.203.91</code>	-v Revealed the same technologies as the basic scan, with additional plugin and response information.
Aggression Level 1	<code>whatweb -a http://10.15.203.91</code>	1 Used minimal requests and produced the same core technology fingerprint as the basic scan
Aggression Level 3	<code>whatweb -a 3 http://10.15.203.91</code>	-v Provided additional information, including DAV/2 confirmation and X-Powered-By header details.

Aggression Level 4	<code>whatweb -a 4 -v http://10.15.203.91</code>	The scan encountered a connectivity error during execution and did not return additional information beyond that obtained at Aggression Level 3.
Follow Redirects	<code>whatweb --follow-redirect=always http://10.15.203.91</code>	No redirects were detected. The results matched those obtained during the basic scan.
Save Results to File	<code>whatweb http://10.15.203.91 whatweb-results.txt -v ></code>	Successfully saved WhatWeb scan results to a text file for documentation and evidence retention.

The WhatWeb scans confirmed that the target web application was hosted on Apache 2.2.8 running on Ubuntu Linux and powered by PHP 5.2.4-2ubuntu5.10. The scans also identified WebDAV Version 2 and the website title "Metasploitable2 - Linux". These findings complemented the results obtained during Nmap HTTP enumeration and provided a more detailed understanding of the technologies deployed on the target web server.

Part 8 - Student Questions

1. What is reconnaissance?

Reconnaissance is the process of gathering information about a target system before conducting further security testing.

2. What is the difference between host discovery and port scanning?

Host discovery identifies which devices are active on a network, while port scanning identifies the open ports and services running on those devices.

3. Explain open, closed and filtered.

- **Open:** A service is running and accepting connections.
- **Closed:** The port is reachable, but no service is listening on it.
- **Filtered:** A firewall or filtering device is blocking the scan, making the port state difficult to determine.

4. What does -sV do?

The -sV option enables service version detection. It identifies the software and version running on open ports.

5. What does -O do?

The -O option performs operating system detection and attempts to identify the target's operating system.

6. Explain what -A combines and why it is noisier.

The -A option combines:

- OS Detection (-O)
- Service Version Detection (-sV)
- NSE Scripts
- Traceroute

It is noisier because it sends more probes and requests to gather additional information from the target.

7. What does -p- mean?

The -p- option tells Nmap to scan all TCP ports from **1 to 65535**.

8. Why can UDP scans take longer?

UDP scans take longer because many UDP services do not respond to requests, so Nmap must wait longer for replies or timeouts.

9. What does -sC do?

The -sC option runs Nmap's default NSE (Nmap Scripting Engine) scripts to gather additional information about services.

10. What information did the HTTP title/headers scripts reveal?

The HTTP scripts revealed information such as:

- The webpage title
- Web server information
- HTTP response headers
- Server software details

11. What did SMB/SSH/FTP enumeration add?

- **SMB Enumeration:** Revealed SMB protocol and sharing information.
- **SSH Enumeration:** Revealed SSH host keys and fingerprints.
- **FTP Enumeration:** Revealed FTP banners and software information.

12. What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap service detection identifies network services and their versions running on open ports. WhatWeb identifies web technologies such as web servers, frameworks, content management systems, and scripting languages.

13. How did WhatWeb levels 1, 3 and 4 differ?

- **Level 1:** Uses minimal requests and provides basic website information.
- **Level 3:** Uses more requests and gathers additional details.
- **Level 4:** Performs the most detailed fingerprinting and may reveal more technologies and configurations.

14. Why must aggressive scanning/fingerprinting remain inside the authorized lab?

Aggressive scanning should only be performed in an authorized lab because scanning systems without permission may violate security policies, organizational rules, or laws. The lab provides a safe environment for learning and testing.

Findings

The reconnaissance activities revealed numerous network services running on the Metasploitable 2 target system. Open TCP services identified during the assessment included:

- FTP (vsFTPD 2.3.4)
- SSH
- Telnet
- SMTP
- DNS
- HTTP
- RPCBind

- SMB
- NetBIOS
- NFS
- MySQL
- VNC
- IRC
- Apache Tomcat
- Java RMI

UDP scanning identified additional services running on the target system, including:

- DNS
- NetBIOS Name Service
- RPCBind
- NFS

Operating system fingerprinting indicated that the target was a Linux-based system, which was consistent with the expected Metasploitable 2 environment.

Nmap Scripting Engine (NSE) enumeration provided additional information regarding:

- HTTP server configuration
- FTP service banners
- SSH host keys
- SMB protocol versions
- Web server headers

Web application fingerprinting was performed using WhatWeb and identified the following technologies:

- Apache 2.2.8
- Ubuntu Linux
- PHP 5.2.4-2ubuntu5.10
- WebDAV Version 2

- Website title: "Metasploitable2 - Linux"

The findings demonstrate that the target system exposes numerous network services and contains multiple legacy technologies commonly associated with intentionally vulnerable training environments. The information collected throughout the assessment was used to create a comprehensive service inventory and provided a clear understanding of the target system's attack surface.

Security Observations

Beyond identifying open ports and services, several findings deserve special attention because they are associated with well-known security risks. These observations highlight technologies and configurations that would typically require remediation in a production environment.

Finding	Why It Matters
vsFTPD 2.3.4 (Port 21)	This version of vsFTPD is historically associated with a publicly documented backdoor vulnerability (CVE-2011-2523). Its presence is a strong indicator of an outdated or intentionally vulnerable system.
Telnet Enabled (Port 23)	Telnet transmits usernames, passwords, and session data in plaintext, making it vulnerable to interception and credential theft. Secure alternatives such as SSH should be used instead.
SMBv1 Enabled (Port 445)	SMBv1 is a legacy file-sharing protocol with multiple known security weaknesses. It has been associated with major attacks and should be replaced with SMBv2 or SMBv3 wherever possible.
Apache 2.2.8 / PHP 5.2.4	Both technologies are end-of-life and no longer receive security updates. Systems running unsupported software are more susceptible to known vulnerabilities and attacks.
rexec and rlogin Services (Ports 512/513)	These legacy remote administration services rely on weak trust-based authentication mechanisms and are considered insecure by modern security standards.
WebDAV Enabled	WebDAV extends HTTP functionality and, if not properly secured, may allow unauthorized file management activities. It can increase the attack surface of a web server.

Finding	Why It Matters
Large Number of Exposed Services	The target system exposes numerous network services including FTP, SSH, Telnet, SMTP, SMB, MySQL, VNC, IRC, and Apache Tomcat. Each additional service increases the attack surface and should be reviewed for necessity.
Multiple Remote Access Services	SSH, Telnet, VNC, rlogin, and rexec were simultaneously available on the target host. Multiple remote access methods increase the number of potential entry points available to an attacker.

The reconnaissance process revealed several intentionally vulnerable and outdated services running on the Metasploitable 2 target system. The presence of legacy technologies such as **vsFTPD 2.3.4**, **Telnet**, **SMBv1**, **Apache 2.2.8**, **PHP 5.2.4**, **rexec**, and **rlogin** demonstrates why these services are commonly used in security training environments and why they would require immediate remediation if discovered on a production system.

Challenges Encountered

- The target system's IP address changed during the exercise following a virtual machine restart (10.15.203.233 → 10.15.203.91). This issue was resolved by verifying the updated IP address using the `ifconfig` command on the Metasploitable 2 system and directing all subsequent scans to the new address. The results obtained before the IP address change remained valid because they were collected from the same target host.
- The installation of WhatWeb initially presented challenges due to package repository and dependency requirements. Several Ruby dependency packages were required during installation. The issue was successfully resolved through package management troubleshooting, allowing WhatWeb to be installed and used for web application fingerprinting.
- The WhatWeb Aggression Level 4 scan did not produce the expected fingerprinting output and instead returned a connectivity error while attempting to access the target web server. This issue was documented during the assessment. The impact was minimal because earlier WhatWeb scans had already successfully identified the key technologies running on the target system, including Apache 2.2.8, PHP 5.2.4-2ubuntu5.10, Ubuntu Linux, and WebDAV Version 2.

FINAL REPORT

The objective of this lab was to perform network reconnaissance and service enumeration against the Metasploitable 2 target system using Nmap and WhatWeb. Various scanning techniques were used to identify active hosts, discover open ports, detect operating system information, enumerate services, and fingerprint web technologies.

The assessment successfully identified several TCP and UDP services, including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, NFS, MySQL, VNC, IRC, and Apache Tomcat. Service version detection, NSE script enumeration, and web fingerprinting provided additional details about the technologies and configurations running on the target system.

The findings confirmed that the target system contained multiple accessible services and legacy technologies commonly found in vulnerable environments. The exercise demonstrated the effectiveness of reconnaissance tools in collecting valuable information and creating a detailed service inventory.

Moreso, the lab objectives were achieved successfully, and the exercise provided practical experience in information gathering, service enumeration, network analysis, and web application fingerprinting within an authorized laboratory environment.

Recommendations

- Disable or upgrade vsFTPD 2.3.4 to a supported and maintained version. Any system still running this version should be considered a high-priority remediation target due to its known security risks.
- Disable Telnet, rsh, rexec, and rlogin services and replace them with SSH for secure remote administration and file transfer.
- Disable SMBv1 and enforce the use of SMBv2 or SMBv3 to improve the security of file-sharing services.
- Upgrade the Apache web server and PHP environment to currently supported and fully patched versions to reduce exposure to known vulnerabilities.
- Review all exposed services, including MySQL, VNC, IRC, Apache Tomcat, and AJP13, and remove or restrict any services that are not required for business or operational purposes.
- Apply the principle of least privilege and restrict access to critical services through appropriate firewall rules, access controls, and network segmentation.

- Conduct regular vulnerability assessments and security updates to ensure that software and services remain secure and properly maintained.